Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Tests

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Abstract

Mutual information (MI) measures how much knowing one variable reduces uncertainty about another. This thesis develops a test for comparing MI between two categorical variables across independent populations. The Normal Wald test uses a normal approximation for this comparison, which can give inaccurate false-positive rates when samples are small or contain many rare category combinations.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains Wald's estimated MI difference and standard error but compares the test statistic with a Student t distribution. To choose its degrees of freedom, the thesis derives a first-order approximation to how much the variance estimate within that standard error changes between samples. The calculation accounts for changes in cell probabilities and their row and column totals.

Simulations assess the approximations and compare false-positive rates and power to detect genuine MI differences. Near independence, sampling noise can make the degrees-of-freedom approximations inaccurate. An online-shopping application illustrates the tests on observed data.

Expanded Welch brings false-positive rates closer to 5% in selected small-sample settings, with lower power under the usual rejection thresholds. We also adjust the rejection thresholds to give both tests approximately 5% false positives at a chosen common MI value. Expanded Welch then has higher power in some comparisons, but this advantage can reverse when a different common MI value is used for the adjustment. Changing the rejection threshold cannot correct bias in the MI difference or errors in its estimated uncertainty.

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Chapter 1

Introduction

Mutual information (MI) measures the average reduction in uncertainty about one variable obtained by observing another (Bossomaier et al., 2016). For example, a categorical input feature X may carry different amounts of information about a class label Y in two populations. This thesis develops and evaluates a statistical test for that comparison using two independently collected samples.

Let P and Q be the joint probability distributions of X and Y in the two populations. The null and alternative hypotheses are

$$H_0 : I(P) = I(Q), \qquad H_1 : I(P) \neq I(Q). \tag{1.1}$$

Under H_0, the populations have equal MI. However, this does not mean that their joint probability tables are the same, as two populations can have equal MI while having different probability tables.

In practice, MI is estimated from observed data. Two samples can produce different MI estimates even when the population MI values are equal, because their observed counts vary from sample to sample. A statistical test must account for this sampling variation before an estimated difference can be interpreted as evidence of a population difference.

1.1 Why the comparison is difficult

The Normal Wald test (Wald, 1943; van der Vaart, 1998) divides the estimated MI difference by its standard error and compares the resulting statistic with a standard normal distribution. At a 5% significance level, the intended false-positive rate is 5% when the population MI values are equal. The normal approximation can be

inaccurate in small or sparse samples, causing the test to reject equal MI too often or too rarely.

Both the MI difference and its standard error are estimated from count tables, whose cells record how often each pair of categories occurs. MI depends on the cell probabilities and their row and column totals. When samples are small and rare combinations of categories occur only a few times or not at all, the resulting probability estimates can distort both the MI difference and its estimated uncertainty.

The proposed Expanded Welch–Satterthwaite test (Expanded Welch) retains Wald's estimated MI difference and standard error but uses a Student t distribution instead of the standard normal distribution. Choosing its degrees of freedom requires estimating how much the MI variance estimates change from sample to sample. For MI, this calculation must account for the fact that changing a cell probability also changes its row and column totals and the MI values calculated from them. This thesis derives how these probability changes affect the estimated MI variance and uses that calculation to obtain degrees of freedom for Expanded Welch.

The Student t distribution gives Expanded Welch a larger rejection cutoff than Normal Wald, so it rejects less often. This can reduce false positives, but it can also reduce statistical power: the probability of rejecting equal MI when the population MI values genuinely differ. The evaluation measures the resulting false-positive rates and nominal power, then uses independent equal-MI simulations to adjust each test's cutoff to target 5% false positives before comparing power. Changing the equal-MI setting used for this calibration checks whether the power comparison is stable. The frequency of usable p-values is reported alongside these rates.

1.2 Research questions

The thesis addresses two research questions:

RQ1: How accurately do the derived approximations predict the mean and sampling variance of the estimated MI variance, and what degrees of freedom do they imply near and away from independence?

RQ2: How do Expanded Welch's false-positive rates and power compare with Normal Wald?

- **RQ2.1:** How do their false-positive rates compare under equal population MI?
- **RQ2.2:** How do their nominal and calibrated power compare when population MI differs?

1.3 Contributions to the field

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance, including the resulting changes in the row and column totals and in pointwise MI.

2. **Methodological contribution.** Expanded Welch uses this variance sensitivity to choose the degrees of freedom for a two-sample test using a Student t distribution.

3. **Empirical contribution.** Controlled simulations check the variance approximations and compare false-positive rates, power and usable results as sample size, table size, category frequencies and MI change. A supporting comparison checks whether sparse-table failures persist under another association construction.

4. **Practical contribution.** The evaluation identifies selected populations at common MI 0.02 nats where Expanded Welch brings false-positive rates closer to 5% than Normal Wald, the dependence of calibrated power on the calibration setting, and estimation errors that make both tests unreliable.

1.4 Scope and organisation

The scope of this thesis is the comparison of MI between two populations using categorical data. Each sample is summarised in a count table, and both tables use the same fixed row and column categories. The two samples are independent, and observations within each sample are also independent. For the main derivation, both populations are assumed to have positive MI and a positive probability in every cell. The case of zero MI, where X and Y are independent, is considered separately. The main simulations compare Normal Wald and Expanded Welch using two-sided tests at the 5% significance level.

Chapter 2 reviews the background and literature. Chapters 3 and 4 define the statistic and derive Expanded Welch. Chapters 5 and 6 present the experiment, results and online-shopping application. Chapter 7 brings together the findings, limitations and future work. The appendices give supporting calculations, experimental settings and additional results.

Chapter 2

Background and Literature Review

Testing for equal MI in two populations requires an estimate from each sample and its sampling variance. This chapter develops the MI estimator, derives its variance and reviews tests for equal population MI.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = -\sum_x p_X(x) \log p_X(x). \tag{2.1}$$

Entropy summarises the uncertainty of one random variable before its value is observed (Shannon, 1948).

For a pair (X, Y), the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X \mid Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y.

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \ldots, r\}$ and $Y \in \{1, \ldots, c\}$. The numbers of categories, r and c, are their alphabet sizes. Write

$$p_{i+} = \sum_{j=1}^{c} p_{ij}, \qquad p_{+j} = \sum_{i=1}^{r} p_{ij} \tag{2.2}$$

for the marginal probabilities, or row and column margins. The pointwise mutual information (also called local MI) associated with cell (i, j) is

$$\ell_P(i, j) = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right). \tag{2.3}$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts, whereas a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^{r}\sum_{j=1}^{c} p_{ij}\ell_P(i, j). \tag{2.4}$$

Here the summations are nested: for each row i, the inner sum runs over all columns j. Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$.

Because P is the joint distribution of (X, Y), we also write $I(P)$ as $I(X; Y)$. Its entropy form is

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \tag{2.5}$$

MI is therefore the average reduction in uncertainty about X obtained by observing Y, or equivalently about Y obtained by observing X. It is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

Although MI measures the overall departure from independence, it does not specify the shape of association. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. The null hypothesis requires equal MI, not identical cell probabilities in the two population tables.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij}. The empirical cell probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}}{n}. \tag{2.6}$$

The plug-in estimator inserts these observed relative frequencies into Equation (2.4):

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij} \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right). \tag{2.7}$$

The zero-cell convention follows from $\lim_{x\downarrow 0} x \log x = 0$, so empty cells contribute no term to the sum. When a positive-probability cell is unobserved, the empirical table assigns it zero probability, contributing to sparse-sample estimation error.

Under multinomial sampling, the cell counts vary between samples but must sum to n. An extra observation in one cell therefore leaves fewer observations for the others. For a fixed set of categories, the plug-in estimate approaches population MI as the sample grows, but its finite-sample bias and sampling variance depend on the complete probability table. In particular, rare cells can change both even when the categories and sample size are unchanged (Paninski, 2003).

Bias is the difference between the average estimate across repeated samples and the population value: $\text{Bias}\{\widehat{I}(P)\} = \text{E}\{\widehat{I}(P)\} - I(P)$. For a fixed $r \times c$ table with positive population cell probabilities, the large-sample expansion has a positive leading term:

$$\text{Bias}\{\widehat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \tag{2.8}$$

The remainder $O(n^{-2})$ collects terms bounded in magnitude by a constant times n^{-2} as n grows for this fixed population table. The leading term grows with table size and shrinks as the sample grows, although it does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing MI estimates from two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2}\left(\frac{1}{n_P} - \frac{1}{n_Q}\right). \tag{2.9}$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MI values. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

Estimation beyond the leading bias correction.

Data are undersampled when too few observations are available to estimate the joint probabilities reliably. Rare category combinations can then be poorly represented or absent from the sample. Panzeri and Treves (1996) derive analytical bias corrections that account for the categories with nonzero population probability, but these remain large-sample approximations. Paninski (2003) shows that substantial error can remain after standard corrections, including the jackknife, when many categories are poorly sampled. Chao and Shen (2003) instead adjust entropy for categories missing from the sample. Their method uses sample coverage, the total population probability of the categories that have been observed, and accounts for the different probabilities of discovering common and rare categories.

Bayesian estimators model the unknown probabilities rather than treating observed proportions as the complete population. Nemenman et al. (2002) combine priors, or probability weights on possible populations, to avoid favouring a narrow range of entropy values before observing the data. This is the Nemenman–Shafee–Bialek (NSB) estimator. More recently, Piga et al. (2024) estimate the strength of a Dirichlet prior from the observed counts. This strength controls how much the probability estimates are drawn towards a uniform distribution, including for unobserved categories.

Entropy estimators can also be combined to estimate MI through $I(X;Y) = H(X) + H(Y) - H(X,Y)$, as in Equation (2.5). Archer et al. (2013) study this route alongside estimators that model the joint probabilities directly. Combining three separately constructed Bayesian entropy estimates need not correspond to one joint probability model and can produce negative MI estimates. Nevertheless, these combined estimators generally outperform the authors' fully Bayesian MI estimator in their tested examples.

The sampling pattern can also suggest a more targeted estimator. Hernández and Samengo (2019) consider a well-sampled, low-entropy variable paired with a variable whose many categories are poorly sampled. They use the well-estimated marginal

and model the conditional probabilities to estimate MI, rather than estimating three entropies separately. Their results also show that combining Bayesian entropy estimates can retain bias under severe undersampling. Estimation errors need not cancel when MI estimates from two distinct populations are subtracted. The two-sample problem therefore requires both a small bias in the estimated difference and a standard error that describes its sampling variation.

2.3 Sampling variance of plug-in MI

Two MI estimates can differ even when their population values are equal, because each estimate varies between samples. Testing their difference therefore requires a standard error that accounts for this sampling variation. We first derive the established large-sample approximation for the variance of one plug-in MI estimate:

$$\mathrm{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n}. \tag{2.10}$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004). Section 2.4 then explains how this result supports a comparison between two populations.

The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_{i} p_{i+} \log p_{i+} - \sum_{j} p_{+j} \log p_{+j}. \tag{2.11}$$

Each sum contains the same function, $f(u) = u \log u$. Expanding this function lets us translate probability errors into changes in MI. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du}\{u \log u\} = \log u + u\frac{1}{u} = \log u + 1. \tag{2.12}$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \tag{2.13}$$

where a is the value around which the function is expanded and $x - a$ is the distance from that value. Here the new value is the estimated probability $x = \widehat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \widehat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\widehat{u} \log \widehat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\widehat{u} - u)}_{\text{first-order change}}. \tag{2.14}$$

Subtracting the population value from both sides makes the change explicit:

$$\widehat{u} \log \widehat{u} - u \log u \approx (\log u + 1)(\widehat{u} - u). \tag{2.15}$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \widehat{I}(P) - I(P) \approx & \sum_{i,j} (\log p_{ij} + 1)(\widehat{p}_{ij} - p_{ij}) \\ & - \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) \\ & - \sum_{j} (\log p_{+j} + 1)(\widehat{p}_{+j} - p_{+j}). \end{aligned} \tag{2.16}$$

The margin errors are sums of the corresponding cell errors:

$$\widehat{p}_{i+} - p_{i+} = \sum_{j} (\widehat{p}_{ij} - p_{ij}), \qquad \widehat{p}_{+j} - p_{+j} = \sum_{i} (\widehat{p}_{ij} - p_{ij}). \tag{2.17}$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_{i} (\log p_{i+} + 1)(\widehat{p}_{i+} - p_{i+}) &= \sum_{i} \left[(\log p_{i+} + 1) \left\{ \sum_{j} (\widehat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.18}$$

The inner sum over j is the probability error for a fixed row i. It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way.

After this rearrangement, the same cell error $\widehat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned}(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ = \log\left(\frac{p_{ij}}{p_{i+}p_{+j}}\right) - 1 \\ = \ell_P(i,j) - 1. \end{aligned} \tag{2.19}$$

Therefore,

$$\widehat{I}(P) - I(P) \approx \sum_{i,j}\{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}). \tag{2.20}$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j}(\widehat{p}_{ij} - p_{ij}) = \sum_{i,j}\widehat{p}_{ij} - \sum_{i,j}p_{ij} = 1 - 1 = 0. \tag{2.21}$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \sum_{i,j}\{\ell_P(i,j) - 1\}(\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j}\ell_P(i,j)(\widehat{p}_{ij} - p_{ij}) - \sum_{i,j}(\widehat{p}_{ij} - p_{ij}) \\ &= \sum_{i,j}\ell_P(i,j)(\widehat{p}_{ij} - p_{ij}). \end{aligned} \tag{2.22}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a. An empirical cell probability is the proportion of observations in that cell, so

$$\widehat{p}_{ij} - p_{ij} = \frac{1}{n}\sum_{a=1}^{n}\left\{\mathbf{1}(Z_a = (i,j)) - p_{ij}\right\}. \tag{2.23}$$

Substituting this expression into Equation (2.22) first gives

$$\widehat{I}(P) - I(P) \approx \sum_{i,j}\ell_P(i,j)\left[\frac{1}{n}\sum_{a=1}^{n}\{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}\right]. \tag{2.24}$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned}\widehat{I}(P) - I(P) &\approx \frac{1}{n}\sum_{a=1}^{n}\left[\sum_{i,j}\ell_P(i,j)\mathbf{1}(Z_a=(i,j)) - \sum_{i,j}p_{ij}\ell_P(i,j)\right] \\ &= \frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\,. \end{aligned} \tag{2.25}$$

The first sum inside the brackets selects the single cell containing Z_a, giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$.

To first order, Equation (2.25) expresses the MI error as an average of n observation-level contributions. The contribution of an observation in cell (i,j) is simply that cell's pointwise MI minus the population average, $\ell_P(i,j) - I(P)$. Its mean is zero because

$$\begin{aligned}\mathrm{E}_P\{\ell_P(X,Y) - I(P)\} &= \sum_{i,j}p_{ij}\{\ell_P(i,j) - I(P)\} \\ &= \underbrace{\sum_{i,j}p_{ij}\ell_P(i,j)}_{I(P)} - I(P)\underbrace{\sum_{i,j}p_{ij}}_{1} = 0. \end{aligned} \tag{2.26}$$

The linear approximation therefore has mean-zero error. The higher-order terms omitted by this approximation generate the leading bias described in Section 2.2. The centred function $\ell_P(i,j) - I(P)$ is called the influence function of MI. Equation (2.25) uses it to approximate the sampling error (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j}p_{ij}\{\ell_P(i,j) - I(P)\}^2. \tag{2.27}$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is

therefore zero. Hence

$$\begin{aligned}\mathrm{Var}\{\widehat{I}(P)\} &\approx \mathrm{Var}\left\{\frac{1}{n}\sum_{a=1}^{n}\{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}\mathrm{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2}\sum_{a=1}^{n}V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \tag{2.28}$$

Averaging n independent contributions therefore divides their variance by n. For dependent observations, covariance terms must also be included in Equation (2.28), as studied by Moddemeijer (1999). The present experiment uses independent observations.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\widehat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0,1). \tag{2.29}$$

This gives a large-sample normal approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i,j) = \log 1 = 0$ in every cell. Hence $I(P) = 0$, $V(P) = 0$, and the first-order average in Equation (2.25) vanishes. A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limiting distribution (Brillinger, 2004; Marinescu and Balcau, 2025). Testing independence therefore uses a chi-squared distribution rather than the normal approximation for comparing positive MI values.

2.4 Approaches to testing MI differences

The preceding section describes the sampling variation of one MI estimate. For two independent samples, their variance contributions add. The square root of an estimate of this sum is the standard error: it estimates the standard deviation of the MI difference across repeated samples. This section reviews the Normal Wald test, the Welch–Satterthwaite principle and other approaches to that comparison. The hypothesis is that the populations have the same MI, $I(P) = I(Q)$, even when their cell probabilities differ.

The Normal Wald test.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with a standard normal distribution (Wald, 1943; van der Vaart, 1998). For MI estimated from independent samples from P and Q, the first-order variance contributions add:

$$\mathrm{Var}\{\widehat{I}(P) - \widehat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \tag{2.30}$$

Each term is the sampling variance of one MI estimate, and independence of the two samples makes their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\widehat{I}(P) - \widehat{I}(Q)}{\sqrt{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q}} \xrightarrow{d} N(0,1) \qquad \text{under } I(P) = I(Q). \tag{2.31}$$

Here $\widehat{V}(P)$ and $\widehat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes positive population cell probabilities, positive pointwise-MI variances and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test. With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses the same standard normal distribution cutoff, regardless of how much the estimated standard error varies.

The Welch–Satterthwaite principle.

This thesis adapts the Welch–Satterthwaite principle from comparing population means to comparing population MI values. Welch's test allows the population variances to differ and compares the standardised difference between two sample means with a Student t distribution with ν degrees of freedom (Welch, 1947). Here $\nu > 0$ controls the distribution's tail weight; an effective value need not be an integer. For two independent samples with sample means $\bar{X}_P, \bar{X}_Q$ and sample variances s_P^2, s_Q^2, the statistic for testing equal population means and the effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \tag{2.32}$$

$$\nu = \frac{\left(s_P^2/n_P + s_Q^2/n_Q\right)^2}{\dfrac{(s_P^2/n_P)^2}{n_P - 1} + \dfrac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \tag{2.33}$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are $n_P - 1$ and $n_Q - 1$. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both samples' degrees of freedom affect ν. Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

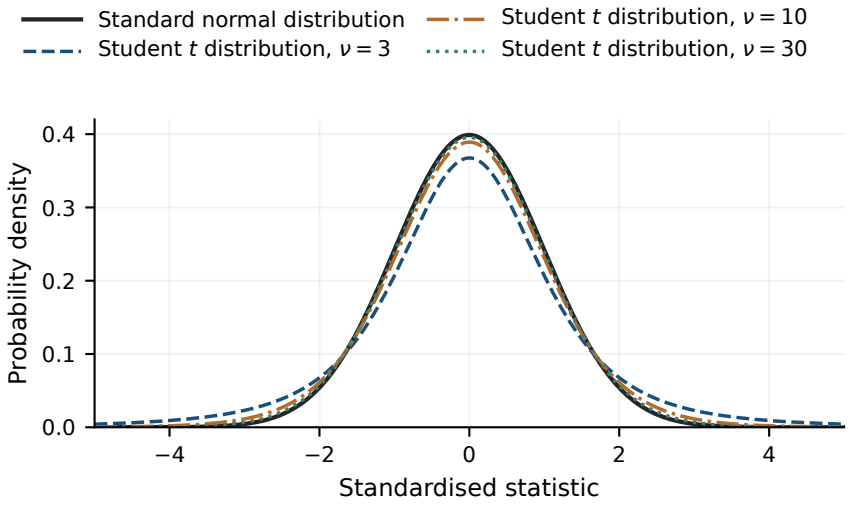


Figure 2.1: Standard normal distribution and Student t distributions with 3, 10 and 30 degrees of freedom.

Figure 2.1 shows that the Student t distribution has heavier tails than the standard normal distribution, especially when ν is small. More probability in the tails means a larger absolute statistic is needed to reject the null at the same significance level. As ν increases, the Student t distribution approaches the standard normal distribution, and their rejection cutoffs become closer.

For MI, the estimated variance contributions are $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. The ordinary $n_P - 1$ and $n_Q - 1$ values follow from the scaled chi-squared distribution of a sample variance under normal sampling; they do not automatically apply to an estimated MI variance. Chapter 4 therefore derives the component degrees of freedom from the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$, using the same moment-matching principle.

The two variance quantities have different roles. $\widehat{V}(P)/n_P$ estimates the sampling variance of $\widehat{I}(P)$, while $\mathrm{Var}\{\widehat{V}(P)\}$ measures how much the variance estimate itself changes between samples. Two variance estimators with the same mean can have different sampling variances. Satterthwaite's calculation matches these two moments, the mean and variance: greater sampling variance relative to the squared mean gives fewer degrees of freedom and a larger rejection cutoff (Satterthwaite,

1946). A sample-size assignment alone does not adapt to differences in this relative variability between probability tables.

The same principle has been used in regression to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). For example, Pan and Wall (2002) calculate approximate degrees of freedom by examining how a regression variance estimate varies between samples, and Expanded Welch uses the variability of the estimated MI variance to choose degrees of freedom in the same way. These regression precedents address variance estimates that are too small or confidence intervals that cover too rarely (Bell and McCaffrey, 2002; Kauermann and Carroll, 2001). A larger cutoff from the Student t distribution can widen the intervals and reduce excessive rejection in such settings.

Previous work on MI uncertainty and comparisons.

Roulston (1999) derives entropy and MI error estimates using error propagation and examines them numerically. These calculations describe the uncertainty of an individual estimate, a necessary input to a comparison of two estimates. The first-order results of Moddemeijer (1989) and Brillinger (2004) give this uncertainty as $V(P)/n_P$, with $V(P)$ determined by the population table. Replacing its probabilities by observed proportions produces the variance estimate used by Wald. This supplies a standard error for the MI difference. Choosing a Welch–Satterthwaite cutoff requires one further quantity: how much the estimated variance varies when a new sample produces a different table.

A Bayesian approach describes uncertainty about population MI given an observed table. Hutter (2002) and Hutter and Zaffalon (2005) model the unknown cell probabilities to calculate the variance and shape of the resulting MI distribution. Our repeated-sampling comparison instead asks how often Normal Wald and Expanded Welch reject equal MI when new samples are drawn from fixed populations P and Q.

Hutcheson's test applies a Welch-type comparison to two estimates of Shannon diversity, or entropy. It uses a Student t distribution with effective degrees of freedom

$$\nu_H = \frac{(s^2_{H,P} + s^2_{H,Q})^2}{(s^2_{H,P})^2/n_P + (s^2_{H,Q})^2/n_Q} \tag{2.34}$$

(Hutcheson, 1970). Here each of $s^2_{H,P}$ and $s^2_{H,Q}$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Hutcheson's rule assigns degrees of freedom from sample

sizes. Expanded Welch instead calculates how the estimated MI variance changes with the table, including the effects of a cell change on its row and column totals.

The entropy comparison establishes that a Student t distribution can be used to compare estimated information quantities, but MI requires more than substituting a different numerator. MI combines the row, column and joint entropies of the same observations. Their estimation errors are connected, because one cell count contributes to all three. The MI variance formulas in Section 2.3 account for that connection. Expanded Welch also tracks how the estimated MI variance changes: a cell change alters both the weights and the pointwise-MI values whose weighted variance is being estimated. Thus Hutcheson supplies a close comparison framework, while the cell and margin derivatives supply the MI-specific inputs.

In a segregation application, Mora and Ruiz-Castillo (2009) derive a normal approximation for comparing positive population MI between populations and time periods, providing a direct precedent for the Normal Wald baseline. Their later peer-reviewed account places the MI index within the segregation literature (Mora and Ruiz-Castillo, 2011).

By contrast, Guha and Chothia (2014) test whether two continuous distributions are identical using MI between pooled observations and their sample labels. For discrete data, Drake and Guha (2014) develop an MI-based k-sample test for equality of distributions. These tests compare complete distributions, not the MI of two variables within each population.

Resampling alternatives.

A bootstrap approximates sampling variation by repeatedly drawing samples with replacement from the observed data (Efron, 1979). For this comparison, samples would be drawn separately from the two observed tables, using their original sample sizes, and the MI difference recalculated for each pair. Resampling from separate empirical probability tables allows their cell probabilities to differ and provides an estimated sampling distribution of the MI difference. The bootstrap variation around the observed estimate could then be used to construct an uncertainty interval for the population difference and assess whether that interval includes zero.

An ordinary empirical bootstrap cannot sample an unobserved cell, because it assigns that cell zero probability. It can therefore miss error arising from unobserved cells. Paninski (2003) gives a broader warning: confidence intervals based on resampling can be too narrow when information estimates remain biased.

A simple permutation test instead combines the two samples and shuffles their group labels. This gives an exact test when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$: the tables can

differ while their MI values are equal. Equal MI alone does not guarantee that this label-permutation test has the stated false-positive rate (Chung and Romano, 2013). Specialised permutation methods allow distinct populations in some parameter comparisons (Janssen, 1997; Chung and Romano, 2013); their validity for equal MI would need to be established.

2.5 Previous work and this thesis's contribution

Existing work provides the leading bias and sampling variance of MI (Miller, 1955; Moddemeijer, 1989; Brillinger, 2004; Paninski, 2003). Hutcheson's entropy comparison assigns degrees of freedom from sample sizes (Hutcheson, 1970), while Pan and Wall (2002) use the variability of a regression variance estimate to choose degrees of freedom. Applying this approach to MI requires calculating how the estimated MI variance changes with the observed table, including changes to the cell probabilities, row and column totals, and pointwise MI values.

This thesis derives how changes in the observed count table affect the estimated MI variance and uses that result to calculate MI-specific degrees of freedom for Expanded Welch. The resulting test retains Normal Wald's MI difference and standard error but compares the statistic with a Student t distribution rather than a standard normal distribution.

The experiment measures errors in the shared MI difference and standard error alongside false-positive rates and power. Paninski (2003) shows that finite-sample information estimates can retain substantial bias after a leading correction. Such bias or an inaccurate first-order variance approximation can undermine both tests, even when Expanded Welch lowers rejection.

Chapter 3

Overview of the Two-Sample MI Test and Wald Baseline

Normal Wald and Expanded Welch share an estimated MI difference and standard error. This chapter defines their statistic and the Normal Wald p-value.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \qquad N^Q = (N_{ij}^Q) \tag{3.1}$$

be independent $r \times c$ count tables with totals n_P and n_Q. The rows and columns represent the same categorical variables X and Y in both populations, with the same category definitions. The empirical probabilities are

$$\widehat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \qquad \widehat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \tag{3.2}$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \tag{3.3}$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \tag{3.4}$$

The large-sample argument uses the following assumptions.

Assumption 1 (Sampling)**.** *Observations are independent and identically distributed within each population, and the two population samples are independent.*

Assumption 2 (Aligned fixed alphabets)**.** *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive cell probabilities)**.** *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

Assumption 4 (Positive MI)**.** *Both populations have positive MI: $I(P) > 0$ and $I(Q) > 0$. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances $V(P) > 0$ and $V(Q) > 0$. It excludes exact independence, where the first-order approximation breaks down.*

Assumption 5 (Logarithm units)**.** *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\widehat{\ell}_P(i,j) = \log\left(\frac{\widehat{p}_{ij}}{\widehat{p}_{i+}\widehat{p}_{+j}}\right) \qquad \text{when } \widehat{p}_{ij} > 0. \tag{3.5}$$

The plug-in MI estimate is

$$\widehat{I}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\widehat{\ell}_P(i,j). \tag{3.6}$$

The same calculation gives $\widehat{I}(Q)$.

The leading fixed-support MI bias appears in Equation (2.8) (Miller, 1955; Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\widehat{\Delta}_{\mathrm{BC}} = \left\{\widehat{I}(P) - \frac{(r-1)(c-1)}{2n_P}\right\} - \left\{\widehat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q}\right\}. \tag{3.7}$$

The two terms in braces are the bias-corrected MI estimates for the separate populations. The correction uses the declared row and column categories because every population cell has positive probability under the model, even when some cells are unobserved in a sample. Corrected estimates can be negative; those values are retained when calculating the difference. When $n_P = n_Q$, the two corrections cancel, so $\widehat{\Delta}_{\mathrm{BC}}$ is exactly the raw plug-in MI difference. With unequal sample sizes,

their difference removes the leading unequal-sample bias described in Section 2.2. A correction that estimates support from occupied cells would be a different estimator.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\widehat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\widehat{V}(P) = \sum_{i,j:\widehat{p}_{ij}>0} \widehat{p}_{ij}\{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2. \tag{3.8}$$

Define $\widehat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\mathrm{SE}}^2 = \frac{\widehat{V}(P)}{n_P} + \frac{\widehat{V}(Q)}{n_Q}. \tag{3.9}$$

Taking the positive square root gives $\widehat{\mathrm{SE}}$, which estimates the standard deviation of $\widehat{\Delta}_{\mathrm{BC}}$ across repeated samples. Table 3.1 distinguishes the variation of pointwise MI across observation outcomes from the sampling variation of the estimates calculated from whole tables.

Table 3.1: Variance quantities in the two-sample MI test.

Quantity	Meaning
$V(P)$	Population variance of pointwise MI; $V(P)/n_P$ approximates $\mathrm{Var}\{\widehat{I}(P)\}$.
$\widehat{V}(P)$	Estimate of $V(P)$ from one sampled count table.
$\widehat{V}(P)/n_P$	Estimated sampling variance of $\widehat{I}(P)$; one contribution to $\widehat{\mathrm{SE}}^2$.
$\mathrm{Var}\{\widehat{V}(P)\}$	Sampling variance of $\widehat{V}(P)$ across repeated samples; used to choose the component degrees of freedom.
$\widehat{\mathrm{SE}}^2$	Estimate of $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, calculated as $\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q$.
$\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$	Sampling variance of $\widehat{\Delta}_{\mathrm{BC}}$; the quantity that $\widehat{\mathrm{SE}}^2$ estimates.

The common standardised statistic is

$$T = \frac{\widehat{\Delta}_{\mathrm{BC}}}{\widehat{\mathrm{SE}}}. \tag{3.10}$$

The sign of T indicates whether P or Q has the larger corrected MI estimate, while its magnitude measures the difference relative to its estimated standard error. Under H_0, we compare T with a reference distribution centred at zero. Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.3 Normal Wald test and motivation for Expanded Welch

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0,1) \qquad \text{under } H_0. \tag{3.11}$$

The two-sided Normal Wald p-value is

$$p_{\mathrm{Wald}} = 2\{1 - \Phi(|T|)\}, \tag{3.12}$$

where Φ is the standard normal distribution function. Under this reference distribution for equal MI, the p-value is the probability of a statistic at least as far from zero as the observed T. The upper-tail probability is $1 - \Phi(|T|)$; symmetry gives the same probability in the lower tail, so the two-sided p-value doubles it. At the nominal 5% level, the test rejects when $p_{\mathrm{Wald}} \leq 0.05$.

Why use a Student t distribution?

Wald uses the standard normal distribution at every sample size, even though its standard error is estimated from the observed tables. Expanded Welch uses the sampling variability of $\widehat{V}(P)$ and $\widehat{V}(Q)$ to choose the degrees of freedom of a Student t distribution. Hutcheson's entropy test and ordinary Welch's test set each variance component's degrees of freedom from its sample size: n_i (Hutcheson, 1970) or $n_i - 1$ (Welch, 1947). Neither assignment measures how cell probabilities and margins make the estimated MI variance change between samples.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For P, to first order,

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P). \tag{3.13}$$

Chapter 4 derives $\mathrm{Var}\{\widehat{V}(P)\}$ by tracking how the calculated MI variance changes when one observation changes the joint table.

Chapter 4

Detailed Derivation of the Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3, but changes the distribution used to judge that statistic from the standard normal distribution to a Student t distribution. To choose its degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples. The name Expanded Welch refers to this expansion of the MI variance as the table probabilities change.

4.1 Choosing degrees of freedom for Expanded Welch

The squared standard error contains two estimated variance contributions, $\widehat{V}(P)/n_P$ and $\widehat{V}(Q)/n_Q$. We assign component degrees of freedom to each contribution, then combine them to choose the test's Student t distribution. For component inputs ν_P and ν_Q, the Welch–Satterthwaite equation is (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\nu_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\nu_Q}. \tag{4.1}$$

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\widehat{V}(P)$. Second, we calculate how $V(P)$ changes when a small amount of probability moves into one cell. Third, that change tells us how

variable $\widehat{V}(P)$ is. Finally, we combine the degrees of freedom from P and Q using Equation (4.1).

The role of the variance estimate follows from the standard construction of a Student t variable. If Z has a standard normal distribution and U independently has a chi-squared distribution with ν degrees of freedom, the scaled ratio follows a Student t distribution:

$$\frac{Z}{\sqrt{U/\nu}} \sim t_\nu. \tag{4.2}$$

Dividing by the independent scaled chi-squared term gives the heavier tails of the Student t distribution. For MI, we approximate this construction by modelling the estimated variance with a scaled chi-squared variable whose mean and variance match its sampling moments. The exact identity assumes the stated normal distribution and independence; the MI reference is an approximation.

Moment matching for one MI variance estimate.

Let $\nu_{P,\text{mom}}$ denote the component target obtained by matching the mean and variance of $\widehat{V}(P)$. Satterthwaite approximates this nonnegative estimate by the scaled chi-squared variable

$$\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\text{mom}}}}{\nu_{P,\text{mom}}}. \tag{4.3}$$

For a chi-squared variable with $\nu_{P,\text{mom}}$ degrees of freedom,

$$\mathrm{E}(\chi^2_{\nu_{P,\text{mom}}}) = \nu_{P,\text{mom}}, \qquad \mathrm{Var}(\chi^2_{\nu_{P,\text{mom}}}) = 2\nu_{P,\text{mom}}. \tag{4.4}$$

The scaling in Equation (4.3) therefore matches the mean automatically:

$$\mathrm{E}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\text{mom}}}}{\nu_{P,\text{mom}}}\right] = \mathrm{E}\{\widehat{V}(P)\}. \tag{4.5}$$

Its variance is

$$\mathrm{Var}\left[\mathrm{E}\{\widehat{V}(P)\}\frac{\chi^2_{\nu_{P,\text{mom}}}}{\nu_{P,\text{mom}}}\right] = \frac{[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu^2_{P,\text{mom}}}\,\mathrm{Var}(\chi^2_{\nu_{P,\text{mom}}}) \tag{4.6}$$

$$= \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\nu_{P,\text{mom}}}. \tag{4.7}$$

Matching this quantity to $\mathrm{Var}\{\widehat{V}(P)\}$ gives

$$\boxed{\nu_{P,\text{mom}} = \frac{2[\mathrm{E}\{\widehat{V}(P)\}]^2}{\mathrm{Var}\{\widehat{V}(P)\}}}. \tag{4.8}$$

The target squares the sampling mean of $\widehat{V}(P)$ and divides by its sampling variance. Dividing $\widehat{V}(P)$ by fixed n_P rescales both the numerator and denominator by $1/n_P^2$, so the same target applies to $\widehat{V}(P)/n_P$ in the squared standard error. To first order, $\mathrm{E}\{\widehat{V}(P)\} \approx V(P)$, so the remaining task is to calculate $\mathrm{Var}\{\widehat{V}(P)\}$.

4.2 How the MI variance changes with the table

To find how much the estimated MI variance changes between samples, we first ask how the population variance responds when a small amount of probability is moved into one cell (Hampel, 1974; van der Vaart, 1998). Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1-\varepsilon)P + \varepsilon\delta_{(x,y)}, \tag{4.9}$$

where $\delta_{(x,y)}$ places probability one on cell (x, y). At $\varepsilon = 0$, the distribution is P. Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one. This direction represents the table error contributed by an observation in cell (x, y). The empirical table averages such one-observation tables, so averaging their first-order effects will give a first-order approximation to the sampling error of $\widehat{V}(P)$. Section 4.3 shows that averaging step after we calculate the effect for one cell.

For an arbitrary cell (i, j),

$$p_{ij}(\varepsilon) = (1-\varepsilon)p_{ij} + \varepsilon\mathbf{1}\{i = x, j = y\}. \tag{4.10}$$

The joint and marginal probability derivatives at the original distribution are

$$\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \tag{4.11}$$

$$\left.\frac{\mathrm{d}p_{i+}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \tag{4.12}$$

$$\left.\frac{\mathrm{d}p_{+j}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \tag{4.13}$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j}\{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \tag{4.14}$$

Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i,j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \tag{4.15}$$

For each logarithm, the derivative is the probability derivative divided by the probability. Substituting the joint and marginal derivatives gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\ell_{P_\varepsilon}(i,j)\right|_0 &= \frac{\mathbf{1}\{i=x, j=y\} - p_{ij}}{p_{ij}} - \frac{\mathbf{1}\{i=x\} - p_{i+}}{p_{i+}} - \frac{\mathbf{1}\{j=y\} - p_{+j}}{p_{+j}} \\ &= \frac{\mathbf{1}\{i=x, j=y\}}{p_{ij}} - \frac{\mathbf{1}\{i=x\}}{p_{i+}} - \frac{\mathbf{1}\{j=y\}}{p_{+j}} + 1. \end{aligned} \tag{4.16}$$

The joint term supplies -1, while subtracting the row and column terms supplies $+1$ each. Their sum is the final $+1$.

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j). \tag{4.17}$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned}\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 &= \sum_{i,j}\{\mathbf{1}(i=x, j=y) - p_{ij}\}\ell_P(i,j) \\ &\quad + \sum_{i,j} p_{ij}\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0. \end{aligned} \tag{4.18}$$

The first sum is $\ell_P(x,y) - I(P)$. In the second sum, the selected-cell term contributes $p_{xy}/p_{xy} = 1$. The row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 = \ell_P(x,y) - I(P)}. \tag{4.19}$$

This is the centred pointwise-MI influence function from Section 2.3.

Effect on the MI variance.

The target is the first-order change in

$$V(P) = \mathrm{Var}_P\{\ell_P(X,Y)\}. \tag{4.20}$$

Write this variance in second-moment form:

$$V(P) = \mathrm{E}_P\{\ell_P(X,Y)^2\} - I(P)^2. \tag{4.21}$$

Define the observation-level variance sensitivity by

$$g_P(x,y) = \left.\frac{\mathrm{d}V(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.22}$$

Differentiating Equation (4.21) gives

$$g_P(x,y) = \left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 - 2I(P)\left.\frac{\mathrm{d}I(P_\varepsilon)}{\mathrm{d}\varepsilon}\right|_0. \tag{4.23}$$

It remains to differentiate

$$\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon)\ell_{P_\varepsilon}(i,j)^2. \tag{4.24}$$

The product rule gives

$$\begin{aligned}\left.\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\,\mathrm{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X,Y)^2\}\right|_0 &= \sum_{i,j}\left.\frac{\mathrm{d}p_{ij}(\varepsilon)}{\mathrm{d}\varepsilon}\right|_0 \ell_P(i,j)^2 \\ &\quad + 2\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0.\end{aligned} \tag{4.25}$$

The first sum is

$$\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\}. \tag{4.26}$$

For the second sum, define the probability-weighted row and column means

$$\mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} = \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}}, \tag{4.27}$$

$$\mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} = \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}}. \tag{4.28}$$

Substitution of Equation (4.16) gives

$$\begin{aligned}&\sum_{i,j} p_{ij}\ell_P(i,j)\left.\frac{\mathrm{d}\ell_{P_\varepsilon}(i,j)}{\mathrm{d}\varepsilon}\right|_0 \\ &\quad = \ell_P(x,y) - \frac{\sum_j p_{xj}\ell_P(x,j)}{p_{x+}} - \frac{\sum_i p_{iy}\ell_P(i,y)}{p_{+y}} + I(P) \\ &\quad = \ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} - \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P).\end{aligned} \tag{4.29}$$

Combining Equations (4.19), (4.23), (4.26), and (4.29) gives the variance sensitivity. To see how its squared term arises, combine the weight-change term in Equation (4.26) with the derivative of $-I(P)^2$:

$$\begin{aligned}
&\ell_P(x,y)^2 - \mathrm{E}_P\{\ell_P(X,Y)^2\} - 2I(P)\{\ell_P(x,y) - I(P)\} \\
&\quad = \ell_P(x,y)^2 - \{V(P) + I(P)^2\} - 2I(P)\ell_P(x,y) + 2I(P)^2 \\
&\quad = \ell_P(x,y)^2 - 2I(P)\ell_P(x,y) + I(P)^2 - V(P) \\
&\quad = \{\ell_P(x,y) - I(P)\}^2 - V(P).
\end{aligned}$$

The second line uses $\mathrm{E}_P\{\ell_P(X,Y)^2\} = V(P) + I(P)^2$; the last line completes the square. Adding twice the expression in Equation (4.29) gives

$$\boxed{\begin{aligned}
g_P(x,y) = \{\ell_P(x,y) - I(P)\}^2 - V(P) \\
+ 2\Big[\ell_P(x,y) - \mathrm{E}_P\{\ell_P(X,Y) \mid X = x\} \\
- \mathrm{E}_P\{\ell_P(X,Y) \mid Y = y\} + I(P)\Big].
\end{aligned}} \tag{4.30}$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for the fact that they change too: each pointwise-MI value uses a cell probability and its row and column totals. The conditional means collect the effects of those changing totals. Both parts are needed to describe how the estimated variance varies across samples.

Centred finite-difference checks on positive tables agreed with the MI and variance derivatives to errors of order 10^{-10}. The experimental supplement gives the checked tables and step size.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \mathrm{Var}_P\{g_P(X,Y)\}. \tag{4.31}$$

The observation-level sensitivities $g_P(x,y)$ have a probability-weighted mean of zero. Averaging the one-cell perturbation directions leaves the population table unchanged:

$$\sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\} = P - P = 0. \tag{4.32}$$

Because the first derivative is linear in the direction of change, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$\mathrm{E}_P\{g_P(X,Y)\} = \sum_{x,y} p_{xy} \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\{(1-\varepsilon)P + \varepsilon\delta_{(x,y)}\}\Big|_0 \tag{4.33}$$

$$= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\}\right\}\Big|_0 \tag{4.34}$$

$$= \frac{\mathrm{d}}{\mathrm{d}\varepsilon} V(P)\Big|_0 = 0. \tag{4.35}$$

Their variation determines how much the estimated MI variance changes between samples.

Assumption 6 (Regular variance sensitivity)**.** *For the first-order Expanded Welch construction,* $\tau^2(P) > 0$ *and* $\tau^2(Q) > 0$.

An observation in cell $Z_a^{(P)}$ can be represented by a table that places probability one in that cell, denoted $\delta_{Z_a^{(P)}}$. Relative to P, that observation contributes the table change $\delta_{Z_a^{(P)}} - P$. The empirical table averages these one-observation tables, so its difference from P is

$$\widehat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \left\{\delta_{Z_a^{(P)}} - P\right\}. \tag{4.36}$$

The plug-in estimate $\widehat{V}(P)$ evaluates V at this empirical table. Equation (4.22) gives its first-order change for one observation's direction. Because a first-order change is linear in the table error (van der Vaart, 1998, Chapter 20), the changes from the observations can be averaged:

$$\widehat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \tag{4.37}$$

The observations are independent, so

$$\mathrm{Var}\{\widehat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \mathrm{Var}_P\{g_P(Z_a^{(P)})\} \tag{4.38}$$

$$= \frac{\tau^2(P)}{n_P}. \tag{4.39}$$

Substituting the first-order mean and variance approximations into Equation (4.8) gives the population first-order approximation to $\nu_{P,\mathrm{mom}}$. Define this value by

$$\boxed{\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)}}. \tag{4.40}$$

To interpret Equation (4.40), divide the sampling variance of $\widehat{V}(P)$ by $V(P)^2$, the square of the population pointwise-MI variance:

$$\frac{\mathrm{Var}\{\widehat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \tag{4.41}$$

so the component degrees of freedom are approximately twice the reciprocal of this relative variance. When the estimate fluctuates little compared with the size of $V(P)$, the degrees of freedom are large and the Student t distribution is close to the standard normal distribution, whereas greater relative fluctuation gives fewer degrees of freedom and a larger rejection cutoff. Positive $V(P)$ does not guarantee positive $\tau^2(P)$: table changes can have no first-order effect on $V(P)$. In that case, describing the sampling variation of $\widehat{V}(P)$ requires a higher-order approximation.

Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.30) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned}\widehat{g}_P(i,j) = {} & \{\widehat{\ell}_P(i,j) - \widehat{I}(P)\}^2 - \widehat{V}(P) \\ & + 2\left[\widehat{\ell}_P(i,j) - \frac{\sum_{j'} \widehat{p}_{ij'}\widehat{\ell}_P(i,j')}{\widehat{p}_{i+}} - \frac{\sum_{i'} \widehat{p}_{i'j}\widehat{\ell}_P(i',j)}{\widehat{p}_{+j}} + \widehat{I}(P)\right].\end{aligned} \tag{4.42}$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical sensitivities also have probability-weighted mean zero under exact substitution. Their variance is centred explicitly:

$$\widehat{\tau}^2(P) = \sum_{i,j} \widehat{p}_{ij}\left\{\widehat{g}_P(i,j) - \sum_{k,l} \widehat{p}_{kl}\widehat{g}_P(k,l)\right\}^2. \tag{4.43}$$

The estimated component degree of freedom is

$$\boxed{\widehat{\nu}_P = \frac{2n_P\widehat{V}(P)^2}{\widehat{\tau}^2(P)}}. \tag{4.44}$$

Repeating the same calculation for the second table gives $\widehat{\nu}_Q$.

Table 4.1 distinguishes the three component quantities. The first two are fixed for a population and sample size; $\widehat{\nu}_P$ changes with each sampled table.

Table 4.1: Component degrees of freedom and their inputs.

Quantity	Role	Inputs
$\nu_{P,\mathrm{mom}}$	Sampling-moment target	$\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{Var}\{\widehat{V}(P)\}$ at fixed P, n_P
$\nu_{P,\mathrm{FO}}$	Population first-order approximation to the target	Population $V(P)$, $\tau^2(P)$ and n_P
$\widehat{\nu}_P$	Component estimate used by Expanded Welch	$\widehat{V}(P)$, $\widehat{\tau}^2(P)$ and n_P from one sample

Combining the two samples' degrees of freedom.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\widehat{\nu}_{\mathrm{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\left\{\widehat{V}(P)/n_P\right\}^2/\widehat{\nu}_P + \left\{\widehat{V}(Q)/n_Q\right\}^2/\widehat{\nu}_Q}}. \tag{4.45}$$

The expanded two-sided p-value is

$$p_{\mathrm{expanded}} = 2\left\{1 - F_t(|T|; \widehat{\nu}_{\mathrm{expanded}})\right\}, \tag{4.46}$$

where $F_t(x;\nu)$ denotes the cumulative distribution function of a Student t distribution with ν degrees of freedom, evaluated at x.

For a fixed finite T, the Student t distribution gives a two-sided p-value at least as large as the standard normal distribution used by Wald. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid. At a common p-value threshold, every Expanded Welch rejection is therefore also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. At the nominal threshold, Expanded Welch cannot raise rejection where Wald is already conservative. Appendix B.4 specifies the componentwise validity rule, including the exclusion of a zero variance component.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \qquad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \tag{4.47}$$

Their totals are $n_P = 105$ and $n_Q = 100$. In N^P, cell $(1,1)$ has count 35, and its row and column each total 50. Its empirical pointwise MI is therefore

$$\widehat{\ell}_P(1,1) = \log\left\{\frac{35/105}{(50/105)(50/105)}\right\} = \log(1.47) \simeq 0.38526.$$

This cell contributes $(35/105)(0.38526)$ to the plug-in MI estimate. Summing the contributions from all nine cells gives $\widehat{I}(P) = 0.13693$; the same calculation for N^Q gives $\widehat{I}(Q) = 0.25848$ nats. Since $(3-1)(3-1) = 4$, the corrected difference is

$$\widehat{\Delta}_{\mathrm{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \tag{4.48}$$

The variance calculation weights each squared deviation from the estimated MI by its cell probability. Cell $(1,1)$, for example, contributes $(35/105)(0.38526 - 0.13693)^2 \simeq 0.02056$ to $\widehat{V}(P)$. Summing over all cells gives $\widehat{V}(P) = 0.27189$ and $\widehat{V}(Q) = 0.43371$. Hence

$$\widehat{\mathrm{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \qquad T = -1.44898. \tag{4.49}$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the same cell illustrates the sensitivity calculation. The first row's pointwise-MI values are 0.38526, -0.51083 and -0.64436, weighted by its counts 35, 10 and 5. Its conditional mean is

$$\frac{35(0.38526) + 10(-0.51083) + 5(-0.64436)}{50} \simeq 0.10308.$$

Because this count table is symmetric, the first column has the same conditional mean. Substituting these values into Equation (4.42) gives

$$\begin{aligned}\widehat{g}_P(1,1) &\simeq (0.38526 - 0.13693)^2 - 0.27189 \\ &\quad + 2\{0.38526 - 0.10308 - 0.10308 + 0.13693\} \\ &\simeq 0.4218.\end{aligned}$$

The probability-weighted mean of the sensitivities is zero, so this cell contributes $(35/105)(0.4218)^2 \simeq 0.0593$ to $\widehat{\tau}^2(P)$. Repeating the calculation for all cells in both tables gives $\widehat{\tau}^2(P) = 0.93212$ and $\widehat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.44) gives component degrees of freedom

$$\widehat{\nu}_P = 16.65, \qquad \widehat{\nu}_Q = 45.86. \tag{4.50}$$

Combining them with Equation (4.45) gives $\widehat{\nu}_{\text{expanded}} = 59.03$ and $p = 0.15264$. Neither method rejects at the 5% level.

Operational algorithm and complexity.

Table 4.2 summarises one comparison.

Table 4.2: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\widehat{V}(P)$, $\widehat{V}(Q)$, the shared standard error, and T.
4	Compute the row and column pointwise-MI means needed in $\widehat{g}_P$ and $\widehat{g}_Q$.
5	Compute $\widehat{\tau}^2(P)$, $\widehat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.45).
7	Return the p-value from the Student t distribution, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$.

4.5 Behaviour and limitations

Near independence.

Consider a population table that approaches independence while its row and column margins stay fixed. At independence, every pointwise-MI value is zero. The linear error in Equation (2.22) weights each cell-probability error by its pointwise MI, so those linear contributions vanish. Products of cell errors can still remain: these are the quadratic terms retained by a second-order expansion. Near independence, the linear weights are small enough that the quadratic terms can contribute substantially.

Along the fixed-margin path, the expansion relates MI, the pointwise-MI variance $V(P)$, and the variance of the sensitivities $\tau^2(P)$. Appendix A.1 gives the calculation and its fixed-margin assumptions. Close to independence on that path, it gives

$$V(P) \approx 2I(P), \qquad \tau^2(P) \approx 4V(P). \tag{4.51}$$

Substituting these local approximations into the component degrees-of-freedom formula gives

$$\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)} \approx \frac{n_P V(P)}{2} \approx n_P I(P). \tag{4.52}$$

The sample-based $\widehat{\nu}_P$ is close to this population first-order value when its estimated inputs are sufficiently accurate. Smaller population MI then gives fewer degrees of freedom and a larger cutoff from the Student t distribution.

Very close to independence, sampling noise itself creates apparent MI in the empirical table. The resulting variance estimates need not be close to their population values, so the estimated degrees of freedom need not tend to zero with population MI at a fixed sample size. These local expansions assume fixed table dimensions, positive cell probabilities and the fixed-margin path. Appendix A.2 retains the quadratic sampling terms needed when population MI is very small.

Theoretical status and validity conditions.

Equations (4.19) and (4.30) are exact derivatives for positive population tables and provide first-order approximations to the sampling errors in MI and its estimated variance. The scaled chi-squared model in Equation (4.3) and the Student t distribution for T are finite-sample approximations. The MI difference and standard error come from the same tables. A sample that changes the estimated difference can also change its standard error, and Expanded Welch's larger cutoff does not account for how the estimated MI difference and its standard error move together. The experimental supplement gives the corresponding first-order covariance calculation. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student t distribution approaches the standard normal distribution used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error, while Expanded Welch also needs positive, finite component and combined degrees of freedom. A failed requirement gives an invalid calculation, not evidence for the null. Appendix B.4 specifies the numerical criteria; the experiment evaluates validity alongside calibration.

Chapter 5

Experimental Design

The simulations draw count tables from populations with known MI. This allows us to measure errors in the estimated MI difference and its standard error, and to count false positives when the population MI values are equal. The evaluation addresses the two research questions in Section 1.2: whether the variance approximations used by Expanded Welch are accurate, and how its false-positive rates and power compare with those of Normal Wald. The design follows the simulation framework of Morris et al. (2019), specifying the populations, quantities being estimated, methods and performance measures.

One observation records an (X, Y) outcome. One simulation repetition draws a sample of observations from each population, producing two count tables to which both tests are applied.

5.1 Setup: Constructing joint distributions with known MI

Before drawing samples, we construct population probability tables with specified MI values. Equal MI values allow us to measure false-positive rates, while different MI values allow us to measure power. For each table, we first choose how frequently the categories occur, then change the association between X and Y while keeping those frequencies fixed.

The main simulations use square tables. Both variables have k categories, labelled $1, \ldots, k$, so the joint distribution has $k \times k$ cells. Let $a = (a_1, \ldots, a_k)$ specify the category frequencies: a_i is the probability of category i, with $a_i > 0$ and $\sum_i a_i = 1$. Within one population, we use this same vector for both variables. Thus a specifies the row margins for X and the column margins for Y.

To obtain independent variables, we could draw X using a, then draw Y independently using a. Cell (i, j) would have probability $a_i a_j$, and MI would be zero. We introduce dependence by allowing Y to copy the category drawn for X on some observations. Let λ be the probability of copying. For each observation, first draw X using a, then choose between two ways of generating Y, independently of the value of X:

- With probability λ, set $Y = X$.
- With probability $1 - \lambda$, draw Y independently using a.

An independent draw can still match X by chance, whereas copying guarantees a match. Increasing λ therefore makes matching categories occur more often.

Both choices preserve the category frequencies of Y. When Y copies X, its categories have probabilities a because those are the probabilities used to draw X. An independent draw of Y also uses a. Consequently, changing how often we copy changes which categories occur together, while the row and column margins remain fixed. Sample row and column proportions fluctuate around these population probabilities.

To calculate a cell probability, add the contributions from the two choices. The independent choice is made with probability $1 - \lambda$ and produces $X = i, Y = j$ with probability $a_i a_j$, contributing $(1 - \lambda)a_i a_j$. The copying choice contributes only to diagonal cells, where the categories match. Reaching cell (i, i) requires both copying and $X = i$, so its contribution is λa_i. Write $\mathbf{1}_{\{i=j\}}$ for an indicator that equals one when $i = j$ and zero otherwise. Adding the two contributions gives

$$p_{ij}(\lambda) = (1 - \lambda)a_i a_j + \lambda a_i \mathbf{1}_{\{i=j\}}, \qquad 0 \leq \lambda < 1, \tag{5.1}$$

where the restriction $\lambda < 1$ keeps the independent choice possible and therefore gives every cell positive probability. This copying construction is a special case of the mixture model described by Bocci et al. (2010).

At $\lambda = 0$, MI is zero. Increasing λ increases MI, although the copying probability needed to reach a particular MI depends on the category frequencies a. To construct a table with the desired MI, we hold a fixed and search numerically for λ, calculating MI from the population table at each candidate value. Appendix B gives the available MI range, the numerical search and the fixed-margin calculation.

Constructing a 2×2 joint distribution with positive MI.

For population P, take $a = (0.7, 0.3)$. Each variable therefore has probability 0.7 of taking category 1 and probability 0.3 of taking category 2. Choose a target MI

of 0.02 nats. The numerical search gives a copying probability of approximately 0.20347.

Cell $(1,1)$ requires $X = 1$, which has probability 0.7. Given $X = 1$, copying always gives $Y = 1$, while an independent draw gives $Y = 1$ with probability 0.7. Cell $(1,2)$ also requires $X = 1$, but copying cannot give $Y = 2$. It therefore requires the independent choice followed by a draw of category 2, which has probability 0.3. Multiplying these conditional probabilities by the probability of $X = 1$ gives

$$p_{11} \simeq 0.7\{0.20347 + (1 - 0.20347)(0.7)\} = 0.53273,$$
$$p_{12} \simeq 0.7(1 - 0.20347)(0.3) = 0.16727.$$

Both off-diagonal cells have probability $(1 - \lambda)(0.7)(0.3)$, so $p_{21} = p_{12}$. The second row totals 0.3, giving $p_{22} \simeq 0.3 - 0.16727 = 0.13273$.

For population Q, use $a = (0.8, 0.2)$ and a target MI of 0.03 nats. Its numerical search gives a copying probability of approximately 0.26091. The resulting tables are

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \qquad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \tag{5.2}$$

These tables have $I(P) = 0.02$ and $I(Q) = 0.03$ nats, so the true signed difference is $\Delta = I(P) - I(Q) = -0.01$ nats. To construct an equal-MI comparison, we instead choose the same target MI for both populations and search for each copying probability separately.

Table sizes and margin regimes.

The experiment repeats this construction for square tables of sizes $2 \times 2, 4 \times 4, 8 \times 8$ and 16×16, containing 4, 16, 64 and 256 cells. Each doubling of the number of categories multiplies the number of cells by four. At a fixed sample size, observations must then be distributed over more possible cells.

At each table size, we compare three patterns of category frequencies, called margin regimes. Uniform margins assign probability $1/k$ to every category. For uneven margins, the first category has probability ρ, and the remaining probability $1 - \rho$ is shared equally among the other $k - 1$ categories. We write this vector as

$$m_k(\rho) = \left(\rho, \frac{1-\rho}{k-1}, \ldots, \frac{1-\rho}{k-1}\right). \tag{5.3}$$

For example, assigning probability 0.8 to the first of four categories gives

$$m_4(0.8) = \left(0.8, \frac{0.2}{3}, \frac{0.2}{3}, \frac{0.2}{3}\right).$$

Table 5.1 specifies the three regimes. Same skew uses the same uneven frequencies in both populations; different skew uses a first-category probability of 0.7 in P and 0.8 in Q. Thus the row and column margins agree within each population, while the margins can differ between populations.

Table 5.1: Row and column margins for each population.

Regime	P: rows and columns	Q: rows and columns
Uniform	$(1/k, \ldots, 1/k)$	$(1/k, \ldots, 1/k)$
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

Under equal MI, the uniform and same-skew regimes give identical joint probability tables, $P = Q$: both populations have the same margins and the same copying probability. The different-skew regime instead gives different tables with equal MI because the category frequencies differ. Comparing the regimes therefore measures false positives under both identical tables and distinct tables with equal MI.

5.2 RQ1: Checking the variance approximations

Expanded Welch chooses its component degrees of freedom from $\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{Var}\{\widehat{V}(P)\}$, and likewise for Q. To check these quantities, we hold the population table P and sample size n_P fixed, then draw repeated count tables. Each draw gives a new $\widehat{V}(P)$, which estimates the pointwise-MI variance $V(P)$. Dividing by n_P gives that population's estimated contribution to the variance of the MI difference. The average of the $\widehat{V}(P)$ values estimates $\mathrm{E}\{\widehat{V}(P)\}$; their variation across draws estimates $\mathrm{Var}\{\widehat{V}(P)\}$.

The observation-level sensitivity derived in Chapter 4 describes how the estimated MI variance responds to changes in the table. Let $\tau^2(P)$ be its population variance. The first-order predictions are

$$\mathrm{E}\{\widehat{V}(P)\} \approx V(P), \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{\tau^2(P)}{n_P}. \tag{5.4}$$

We compare $V(P)$ with the observed mean of $\widehat{V}(P)$, and $\tau^2(P)/n_P$ with its observed sampling variance. We also average the sample-based prediction $\widehat{\tau}^2(P)/n_P$ across the repeated tables. Comparing this average with the same observed sampling variance shows the effect of estimating the sensitivity from each sample.

We next compare the component degrees of freedom with the sampling-moment target from Chapter 4:

$$\nu_{P,\mathrm{mom}} = \frac{2\{\mathrm{E}\,\widehat{V}(P)\}^2}{\mathrm{Var}\{\widehat{V}(P)\}}. \tag{5.5}$$

The repeated samples estimate the mean and variance in this expression. We compare $\nu_{P,\mathrm{FO}}$, calculated from the population probabilities, and the average of the sampled $\widehat{\nu}_P$ values with this target. Averaging $\widehat{\nu}_P$ need not recover $\nu_{P,\mathrm{mom}}$: the estimate is a nonlinear ratio of sample-based quantities, whereas the target uses the sampling mean and variance of $\widehat{V}(P)$. For the sampled estimates, the ratio $\mathrm{mean}(\widehat{\nu}_P)/\nu_{P,\mathrm{mom}}$ equals one when the average estimate matches the target; a ratio below one means that the estimated degrees of freedom are lower on average. Medians and percentile ranges describe their sampling spread. When a summary uses only positive finite degrees of freedom, we report the included fraction. The same checks are made for Q. Appendix B.7 gives the definitions and algebra.

Table 5.2 specifies two grids. The first varies sample size at common MI 0.02 nats. The second includes very small positive MI to examine behaviour near independence, where the first-order variance becomes small and quadratic sampling terms can matter. Both grids use all three margin regimes and equal sample sizes $n_P = n_Q = n$. Each row combines every listed table size, MI value and sample size.

Table 5.2: Settings for the approximation checks and broad null comparison.

Check	Parameter values
Across sample sizes	$k \in \{2, 4, 8, 16\}$, $I(P) = I(Q) = 0.02$ nats. $n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$.
Near independence	$k = 4$, $n \in \{100, 1000, 10000\}$. $I(P) = I(Q)$ takes the values (nats) $0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05$.

5.3 RQ2.1: False-positive rates under equal MI

To measure false positives, we set the two population MI values equal and draw a count table from each population in every repetition. Every rejection is then a false positive. Normal Wald and Expanded Welch use the same bias-corrected MI difference and estimated standard error from Chapter 3. Their cutoffs differ: Normal Wald uses the standard normal distribution, while Expanded Welch uses a Student t distribution with MI-specific degrees of freedom estimated from the two sampled count tables.

Both tests use a two-sided 5% nominal test. The broad null comparison uses the first grid in Table 5.2: common MI 0.02 nats, four table sizes, three margin regimes and ten sample sizes, giving 120 settings. A false-positive rate above 5% means excessive rejection; a rate below 5% means conservative rejection. Lower rates alone do not establish better testing: a rule that never rejects has zero false positives and zero power. We assess nominal calibration by closeness to 5% and evaluate power separately.

We also report the fraction of valid results. Because invalid calculations contribute no rejection to the all-repetition rate, failures can lower that rate without improving the decisions made on usable samples.

To check whether rejection behaviour depends on the association construction, a supporting comparison replaces copying by a probability transfer within four cells while preserving the margins and target MI. The experimental supplement reports the full construction results.

5.4 RQ2.2: Nominal and calibrated power

Power is the probability of rejecting equal MI when the population MI values differ. We first compare power using the nominal 5% cutoffs. A test that rejects less often under equal MI can also reject less often under alternatives, so we then compare power after adjusting the cutoffs to match rejection at a specified equal-MI setting.

The focused comparison uses settings A–C in Table 5.3. Earlier simulations identified A and B as settings where Expanded Welch brought false-positive rates closer to 5%; their power is evaluated on new independent samples. Setting C provides a larger-sample comparison.

To choose the MI differences for a power curve, we first measure the sampling variation under equal MI. An independent null simulation estimates the standard deviation (SD) of the corrected MI difference; denote this fitted value by σ_0. Measuring differences in units of σ_0 makes the three curves comparable relative to their null sampling variation.

Table 5.3: Focus settings for the power comparison; common null MI is 0.02 nats.

Setting	Margins	$n_P = n_Q$	Size	Fitted null SD σ_0 (nats)
A	Uniform	20	4×4	0.154967
B	Different skew	100	8×8	0.065138
C	Different skew	10,000	4×4	0.003175

For each curve, we hold $I(P) = 0.02$ nats and increase $I(Q)$. Let h be the number of fitted null SDs separating the population MI values. The populations satisfy

$$\begin{aligned} I(P) = 0.02, \qquad I(Q) &= 0.02 + h\sigma_0, \\ h &\in \{0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4\}, \\ \Delta = I(P) - I(Q) &= -h\sigma_0. \end{aligned} \tag{5.6}$$

Thus $h = 0$ gives equal MI, while positive h gives larger MI in Q and a negative signed difference. For example, at C, two null SDs are approximately $2(0.003175) = 0.00635$ nats. The corresponding alternative has $I(P) = 0.02$, $I(Q) \simeq 0.02635$ and $\Delta \simeq -0.00635$ nats. The fitted SD and the resulting population tables stay fixed during evaluation. Appendix B gives their construction and the fixed SD values.

Power after matching null rejection.

Empirical calibration chooses a p-value threshold separately for each test. For each full power curve, the procedure is:

1. Use the independent fitting samples with $I(P) = I(Q) = 0.02$ nats, retaining the curve's margins, table size and sample sizes.
2. Choose each rule's threshold to give at most 5% rejection in that fitting batch, and hold it fixed across the curve.
3. Apply the thresholds to separate equal-MI evaluation samples and to the alternative samples, measuring the achieved false-positive rates and power.

The thresholds are fitted for these known simulation settings, where calibrated power compares the tests after matching false-positive rates.

Normal Wald ranks simulated datasets by $|T|$: larger $|T|$ always gives a smaller p-value and stronger evidence against equal MI. Expanded Welch's p-value also depends on degrees of freedom that change with the sampled tables, so it can rank the datasets differently. At a common p-value threshold, its rejections are a subset of Wald's. Calibration fits a separate threshold for each test, so that earlier nesting no longer determines which test has greater power.

Changing the calibration setting.

Equal MI can occur at different common MI values. To examine how that choice affects the power comparison, we repeat calibration at $h = 2$ and $h = 4$ for A–C using the two equal-MI settings in Table 5.4.

The lower null sets both MI values to the alternative's smaller value; the upper null sets both to its larger value. Both retain the alternative's margins and sample sizes.

Table 5.4: The alternative and two calibration nulls; MI values are in nats.

Scenario	$I(P)$	$I(Q)$
Fixed alternative	0.02	$0.02 + h\sigma_0$
Lower null	0.02	0.02
Upper null	$0.02 + h\sigma_0$	$0.02 + h\sigma_0$

Independent upper-null batches fit and validate the new cutoffs. We apply both calibrations to the same alternative simulation outputs, so any change in measured power comes from the cutoffs rather than a change of alternative samples. Each calibration is also evaluated at both null settings. Matching rejection at one equal-MI setting does not ensure the same false-positive rate elsewhere under equal MI. Appendix B.8 gives the construction and uncertainty calculation.

The experimental supplement specifies the broader comparison over MI differences from 0 to 0.2 nats at $n = 1{,}000$ and 10,000, together with a separate sample-allocation analysis.

5.5 Shared evaluation: Sampling and uncertainty

At one evaluation setting, each repetition draws a count table from P and an independent count table from Q. Both tests use these same two tables, so their rejection decisions can be compared on the same sampled data. Each batch contains 100,000 independent repetitions. More repetitions improve the precision of the reported rates; each test still uses only n_P and n_Q observations within a repetition. The study uses three types of batch:

1. **Base evaluation:** 366 settings across the main and supplementary comparisons, measuring rejection and estimation accuracy.
2. **Independent fitting:** the equal-MI populations for A–C, used to estimate the null SDs and fit calibration cutoffs.
3. **Calibration sensitivity:** six upper-null settings, each with separate fitting and validation batches.

Separate random streams supply fitting and evaluation samples. Settings shared between comparisons are sampled once within each role.

Table 5.5 defines the measures for each evaluation setting.

Bias uses finite MI differences even when a test is invalid. The variance ratio uses repetitions with both finite $\hat{\Delta}_{\mathrm{BC}}$ and finite $\widehat{\mathrm{SE}}^2$. A ratio of one indicates agreement.

Table 5.5: Performance measures calculated from repeated samples.

Measure	Calculation
Rejection rate	$\frac{\text{number of valid rejections}}{\text{number of simulation repetitions}}$
Valid-result fraction	$\frac{\text{number of valid outputs}}{\text{number of simulation repetitions}}$
Difference bias	$\text{mean}(\widehat{\Delta}_{\text{BC}}) - \Delta$
Variance ratio	$\frac{\text{mean}(\widehat{\text{SE}}^2)}{\text{observed variance of } \widehat{\Delta}_{\text{BC}}}$
Direction under alternatives	Correct when $\widehat{\Delta}_{\text{BC}}\Delta > 0$; opposite when $\widehat{\Delta}_{\text{BC}}\Delta < 0$.

Ratios below one mean that the squared standard error understates the observed difference variance on average; ratios above one indicate overstatement. Power includes opposite-direction rejections, so we report estimated direction separately. Appendix B.5 specifies the inclusion rules and complete calculations.

Rejection rates have Wilson 95% confidence intervals (Wilson, 1927), which describe uncertainty from the finite number of simulation repetitions. Intervals for method differences account for both methods being applied to the same simulated data. For calibrated power, we resample fitting and evaluation outputs and refit the cutoffs, so the intervals also include cutoff-fitting uncertainty. We compare power-difference intervals with a practical tolerance of ± 1 percentage point. These are pointwise intervals for one rate or comparison at a specified setting. Appendix B gives the interval calculations.

Chapter 6

Results

This chapter checks the approximations used to choose Expanded Welch's degrees of freedom, then compares its false-positive rates and power with Normal Wald. Supporting simulations examine additional settings, and the online-shopping application illustrates interpretation of the test outputs.

One simulation repetition consists of independently drawing one count table from P and one from Q. Normal Wald and Expanded Welch are both applied to the same two sampled tables in that repetition. Each evaluation setting uses 100,000 repetitions.

6.1 RQ1: Accuracy of variance approximations

For a fixed population table P and sample size n_P, each repetition produces a new $\widehat{V}(P)$. We examine its average value, $\mathrm{E}\{\widehat{V}(P)\}$, and its variation between repetitions, $\mathrm{Var}\{\widehat{V}(P)\}$. These are the two moments needed for the component degrees of freedom. Table 3.1 in Chapter 3 distinguishes these quantities from the sampling variance of the MI difference.

6.1.1 Mean and sampling-variance predictions

Equation (5.4) predicts $\mathrm{E}\{\widehat{V}(P)\} \approx V(P)$ and $\mathrm{Var}\{\widehat{V}(P)\} \approx \tau^2(P)/n_P$, where $\tau^2(P)$ measures the variability of the observation-level sensitivities derived in Chapter 4. We compare each prediction with the corresponding moment observed across repetitions, and repeat the checks for Q.

Figure 6.1 shows the different-skew populations at common MI 0.02 nats. The top row divides the simulated mean of $\widehat{V}(P)$ by the population benchmark $V(P)$, and likewise for Q. The middle row divides $\tau^2(P)/n_P$ by the observed $\mathrm{Var}\{\widehat{V}(P)\}$. The

bottom row uses the average $\widehat{\tau}^2(P)/n_P$ as its numerator and the same observed variance as its denominator. Both variance rows use the same vertical scale, so their differences show the effect of substituting sample proportions. A ratio of one indicates agreement. Shading in the first two rows shows approximate 95% Monte Carlo intervals from 100,000 repetitions; their half-widths are below 1.3% of the plotted values, making them barely visible on this scale.

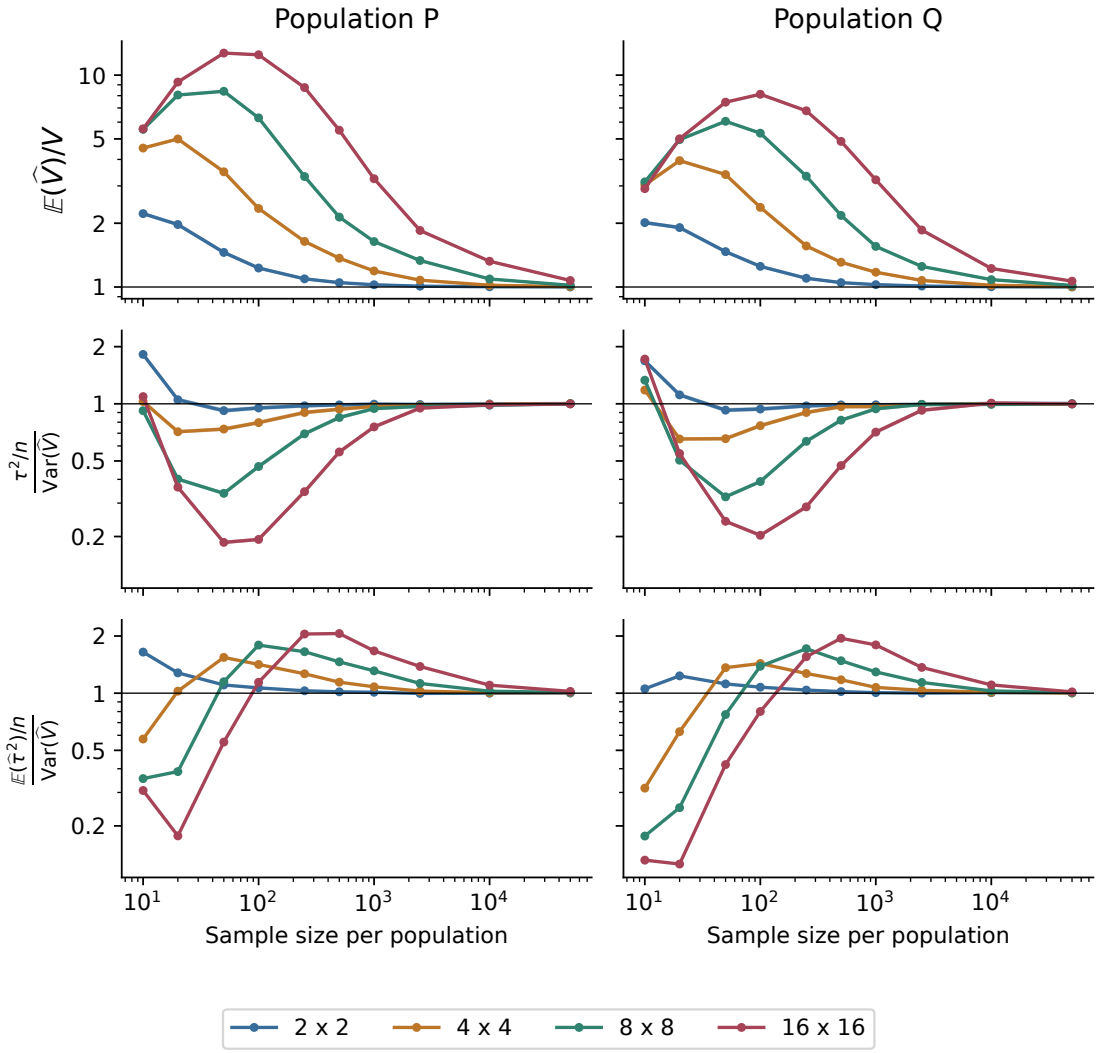


Figure 6.1: Mean and sampling-variance ratios under different-skew margins.

In the top row of Figure 6.1, the larger-table mean ratios first rise and then fall. Initially, many rare combinations are unobserved and contribute nothing to $\widehat{V}$. As more appear, cells observed only once can have large estimated pointwise MI. For example, if a cell and its row and column each contain one observation, its empirical pointwise MI is $\log n$. Such values can be far from the table's plug-in MI and contribute strongly to $\widehat{V}$. A decomposition of the 16×16 sampled tables confirms that once-observed cells dominate its mean around the peak (Appendix C.1).

Their contribution then falls as repeated counts give better estimates of the cell probabilities. Under these margin rules, larger tables spread probability over more rare combinations, extending the sample-size range over which this distortion matters.

In both top panels, the 2×2 mean ratio falls towards one without a pronounced peak over the plotted range; the 4×4 peak is smaller and earlier than those for the two larger tables. The larger tables also take more observations to recover. The normalised mean overshoot is greater in P than in Q, although Q has more uneven margins. Changing the margins changes both the population benchmark V and which rare cells appear in a sample, so equal MI does not give equal relative errors.

The middle-row dips show a related limitation of the linear approximation. The prediction $\tau^2(P)/n_P$ includes the first-order change in $\widehat{V}(P)$, while the empirical log ratios can change substantially when rare cells appear or their counts increase. Checks of the linear change and the remaining error find substantial variation beyond the linear term around the larger-table peaks. The observed sampling variance then exceeds the prediction, putting the ratio below one.

In the bottom row, substituting sample proportions changes the sensitivities used to estimate $\tau^2(P)$. Around the peak, once-observed cells contribute strongly to this estimate as well as to $\widehat{V}(P)$. The sample-based prediction can consequently overshoot the observed sampling variance, giving bottom-row ratios above one even while the population prediction remains below it.

The contrast between the middle and bottom rows is strongest for the larger tables in both populations: their population predictions dip farther below the observed variance, while their sample-based predictions first underestimate it and then overshoot. The 2×2 curves have much smaller departures and settle near one sooner. Thus sample substitution can reverse the direction of the prediction error, rather than simply reduce the error in the population formula. For 16×16, the deepest dip and the transition to sample-based overprediction occur later in Q than in P, showing that margins also affect the timing of these errors.

Across all three rows, larger samples make empirical probabilities more accurate and the variance ratios approach one. The mean ratios can remain above one after the variance ratios are already close, so accurate prediction of variation between samples does not require equally accurate prediction of the mean. The experimental supplement provides the companion uniform and same-skew plots.

6.1.2 How moment errors affect degrees of freedom

Equation (5.5) defines $\nu_{P,\mathrm{mom}}$ from $\mathrm{E}\{\widehat{V}(P)\}$ and $\mathrm{Var}\{\widehat{V}(P)\}$. We compare this target with $\nu_{P,\mathrm{FO}}$, calculated from the population first-order formula, and the average of $\widehat{\nu}_P$, estimated from each sampled table. Table 4.1 identifies their inputs. The first comparison checks the moment approximations; the second also includes the effect of substituting sample proportions.

For the same different-skew populations, Figure 6.2 uses the moment target, estimated from the simulations, as the common denominator. The top row shows $\nu_{P,\mathrm{FO}}/\nu_{P,\mathrm{mom}}$; the bottom row shows $\mathrm{E}(\widehat{\nu}_P)/\nu_{P,\mathrm{mom}}$, and likewise for Q. Both rows use the same vertical scale. Ratios below one indicate fewer degrees of freedom than the moment target; ratios above one indicate more. A bottom-row value is shown only when that component's estimate is positive and finite in all 100,000 repetitions. Shading in the top row gives approximate 95% Monte Carlo intervals.

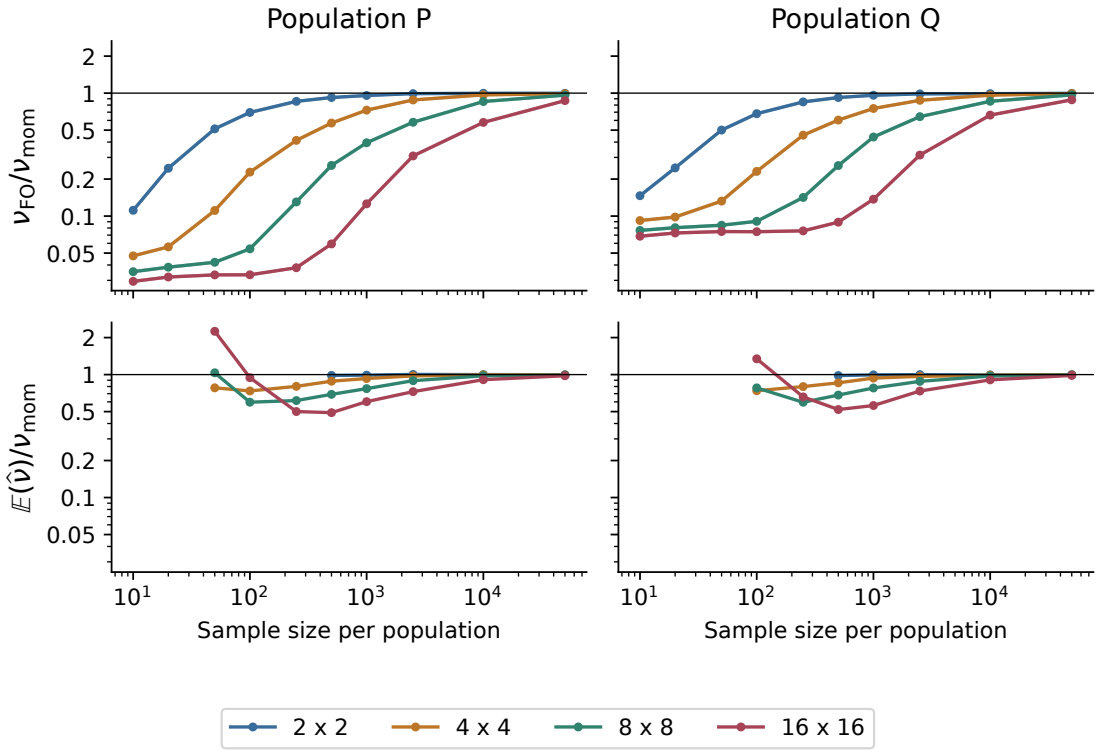


Figure 6.2: Component degrees-of-freedom approximations relative to their moment targets.

The top row of Figure 6.2 combines two moment errors that act in opposite directions. In the moment-matching formula, underestimating $\mathrm{E}\{\widehat{V}(P)\}$ lowers the squared numerator, whereas underestimating $\mathrm{Var}\{\widehat{V}(P)\}$ lowers the denominator and raises the resulting degrees of freedom. The top row therefore reflects both errors in Figure 6.1. Over most of the sample-size range, the mean error dominates and the

first-order degrees of freedom fall below the moment target. As the moment ratios approach one, the first-order value approaches that target.

In both top panels, 2×2 reaches the target first, followed by 4×4; the two larger tables remain below it over a wider sample-size range. This follows the size ordering of the mean errors in Figure 6.1, amplified by squaring the mean. The first-order ratios are generally closer to one for Q, whose normalised mean overshoot is smaller.

For different-skew 4×4 populations at $n = 1{,}000$ and MI 0.02, the sampling-variance prediction is already close to its observed value, but the mean is still underestimated. Squaring that smaller mean leaves the first-order degrees of freedom below the moment target: 15.30 rather than 21.03 for P. Table 6.1 gives the corresponding comparisons for both populations; df abbreviates degrees of freedom. Appendix B.7 gives the algebra.

Table 6.1: Component moment and degrees-of-freedom comparisons at 4×4, $n = 1{,}000$, different skew.

	Moment ratios			Component df		
MI, pop.	Mean ratio	Var. ratio (pop.)	Var. ratio (sample)	Mean df	Moment df	ν_{FO}
0.0001, P	44.939	0.047	1.965	4.86	9.12	0.10
0.0001, Q	41.996	0.052	2.142	4.44	8.63	0.09
0.02, P	1.190	0.970	1.080	19.55	21.03	15.30
0.02, Q	1.174	0.967	1.072	17.00	18.14	13.61
0.05, P	1.066	1.008	1.013	42.56	43.07	37.63
0.05, Q	1.057	1.031	1.029	37.59	38.61	33.50

In Table 6.1, "Mean ratio" divides the simulated mean of $\widehat{V}(P)$ by $V(P)$. Both "Var. ratio" columns divide their prediction by the observed $\mathrm{Var}\{\widehat{V}(P)\}$: "pop." uses the population prediction; "sample" averages the sample-based predictions.

For component degrees of freedom, "Mean df" averages $\widehat{\nu}_P$, "Moment df" gives $\nu_{P,\mathrm{mom}}$, and ν_{FO} gives $\nu_{P,\mathrm{FO}}$. The corresponding definitions apply to Q. All 100,000 component estimates are positive, finite and included in each row.

Returning to the bottom row of Figure 6.2, sample substitution changes both parts of the degrees-of-freedom formula. Using $\widehat{V}(P)$ in the numerator captures its enlarged sampling mean, but the sample-based prediction of its sampling variance can be too large.

These bottom panels also recover in table-size order, but their early curves differ from the broadly rising first-order ratios. The first shown 16×16 estimates exceed

the target, then fall below it as the sample-based variance prediction changes from too small to too large. The 4×4 and 8×8 estimates have smaller excursions before recovering. The 2×2 summaries begin later and are already close to the target when shown. Their missing earlier points reflect the requirement that every component estimate be positive and finite; they do not represent zero degrees of freedom.

For a given numerator, a larger variance prediction gives fewer degrees of freedom. Averaging the sampled ratio also affects the size of the error; it is not equivalent to inserting average inputs into the formula.

At larger sample sizes in these bottom panels, the sample-based variance prediction improves and the mean estimated degrees of freedom recover towards the moment target. The population first-order value can remain farther below the target because it still uses $V(P)$ in place of the enlarged sampling mean. Table C.2 gives means, medians and sampling ranges at the focus and sparse-table settings; all 100,000 component estimates are included in each row. The moment target itself does not guarantee an exact Student t distribution for the test statistic.

6.1.3 Behaviour near independence

Near independence, the pointwise-MI weights in the linear sampling error become small, so quadratic terms in the cell-probability errors can contribute substantially to the variability of the estimates. Even when population MI is almost zero, sampling can produce apparent association and positive $\widehat{V}(P)$ and $\widehat{V}(Q)$ values. The near-independence calculation in Appendix A.2 retains these quadratic terms to predict the moments of $\widehat{V}$ and the mean estimated component degrees of freedom.

Figure 6.3 shows 4×4 different-skew populations at $n = 1{,}000$. The first two rows compare the observed mean and sampling variance of $\widehat{V}$, respectively, with their near-independence predictions. The bottom row compares the mean estimated degrees of freedom, its near-independence prediction, the sampling-moment target and the population first-order value.

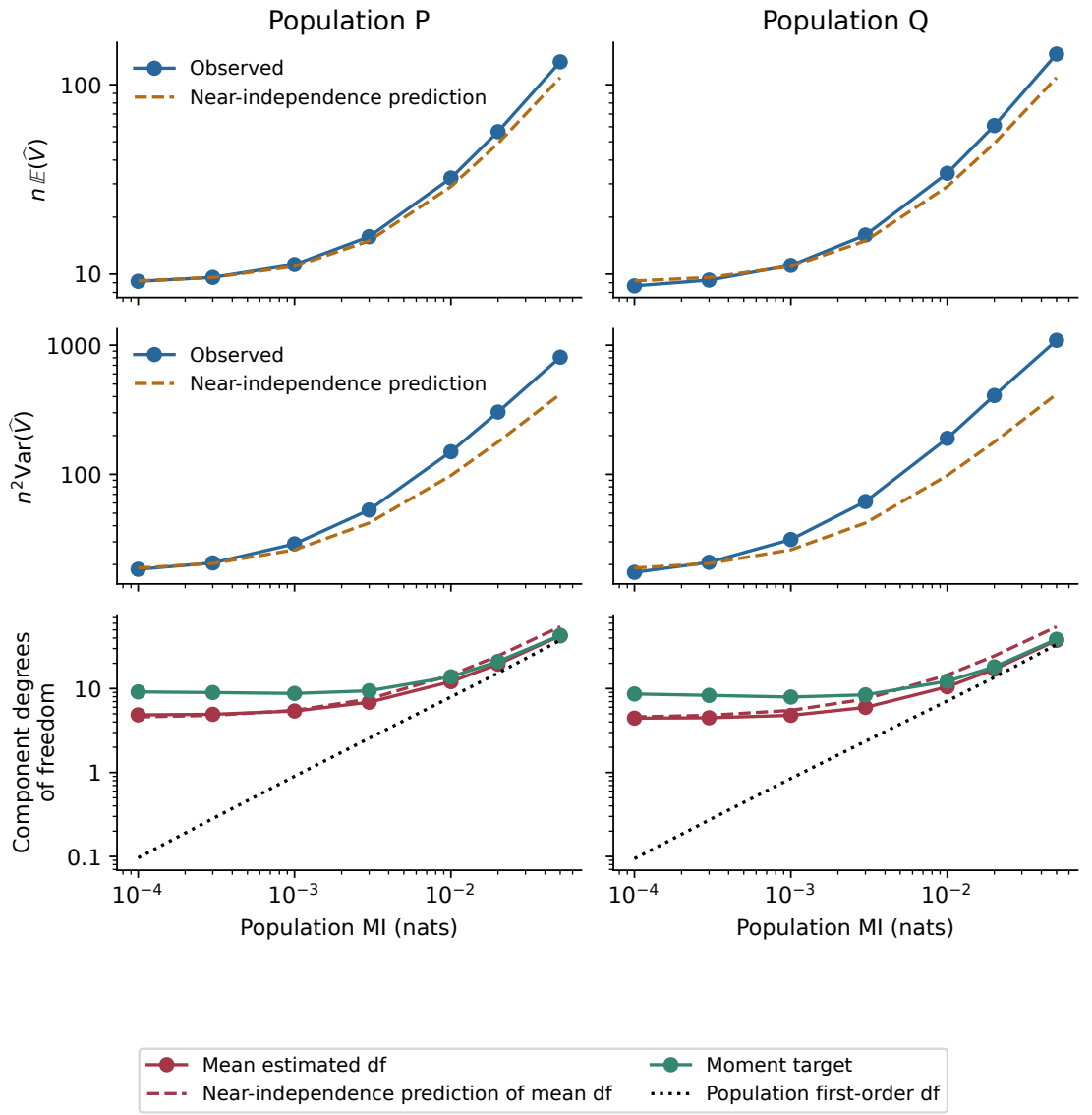


Figure 6.3: Component moments and degrees of freedom near independence.

At the smallest MI values in Figure 6.3, the top and middle rows show that the near-independence predictions capture the positive mean and sampling variance of $\widehat{V}$ sustained by sampling noise. In the bottom row, the quadratic sampling contribution also explains why the mean estimated component degrees of freedom remain several units, around half the moment target. The population first-order degrees of freedom approach zero because they omit that quadratic contribution.

In the bottom row, the moment target has a shallow initial dip before rising. Across those early MI settings, the observed sampling variance grows faster than the square of the observed mean, lowering their moment-matching ratio. At larger MI the squared mean grows faster and the target rises. The mean estimated degrees of freedom also rise towards the target, so their initial factor-of-two separation does not persist across the whole MI range.

Across the figure, stronger population association raises the observed moments and degrees of freedom as MI increases. The dashed predictions become less accurate as the terms omitted from the local expansion around independence become more important. In particular, the approximation $\widehat{V}(P) \approx 2\widehat{I}(P)$ increasingly understates the observed mean and sampling variance of $\widehat{V}(P)$, with a similar pattern for Q. The middle-row separation therefore reflects a loss of accuracy in the local sampling-variance prediction. Appendix C.1 gives the supporting checks. The experimental supplement retains the full diagnostic plots.

Comparing the two columns, the local moment predictions are the same for P and Q, since they use the same table size, sample size and MI. The observed moments separate as MI increases, with a larger sampling variance in Q. The different margins therefore matter beyond the quantities retained in this local approximation. The discrepancy is greater in the sampling-variance row than in the mean row in both columns.

In the bottom row, the near-independence prediction rises above the mean estimated degrees of freedom as MI increases, while the population first-order curve remains below the moment target. The local substitutions $\widehat{V} \approx 2\widehat{I}$ and $\widehat{\tau}^2 \approx 4\widehat{V}$ increasingly understate the sampling means of these inputs. Their combined error in the nonlinear degrees-of-freedom formula gives an upward error in the predicted mean here, even though the local moment predictions in the upper rows are too small.

Table 6.1 compares two other predictions of $\mathrm{Var}\{\widehat{V}(P)\}$: the population first-order value $\tau^2(P)/n_P$ and the average sample-based value $\widehat{\tau}^2(P)/n_P$. The same comparisons apply to Q. At the smallest MI target in Figure 6.3, the population first-order predictions are only about 5% of the observed sampling variances. The sample-based predictions average roughly twice the observed sampling variances. This upward error follows from the local sample-based relation $\widehat{\tau}^2(P) \approx 4\widehat{V}(P)$: sampling noise contributes to both estimates, producing an inflated prediction of variation (Appendix A.2).

At MI 0.02, the population first-order predictions of $\mathrm{Var}\{\widehat{V}(P)\}$ and $\mathrm{Var}\{\widehat{V}(Q)\}$ are much more accurate. The sample-based predictions still overestimate those sampling variances, with ratios above one for both populations. The population first-order variance calculation improves as association strengthens, while the local near-independence calculation loses accuracy. Underestimating the mean of $\widehat{V}(P)$ still lowers $\nu_{P,\mathrm{FO}}$ relative to $\nu_{P,\mathrm{mom}}$, and likewise for Q. Appendix C.1 reports the mean error across MI and margins.

Table 6.1 shows the continuation to MI 0.05: all three moment ratios are close to one, and the mean estimated degrees of freedom are closer to the moment target than

the population first-order values. Across the MI levels, Q has slightly smaller mean ratios but also lower absolute component degrees of freedom. Its sampling variance is greater relative to its squared sampling mean, giving a lower moment target. Relative accuracy and the absolute size of the degrees of freedom are therefore different comparisons.

6.2 RQ2.1: False-positive rates under equal MI

Under $I(P) = I(Q)$, every rejection is a false positive. Both tests use $T = \widehat{\Delta}_{\mathrm{BC}}/\widehat{\mathrm{SE}}$, but Expanded Welch uses a cutoff from the Student t distribution in place of Wald's standard normal cutoff. At the common nominal p-value threshold, the larger cutoff reduces rejection. We assess whether that reduction brings the false-positive rate closer to the nominal 5% level, and examine the shared estimates to interpret the observed patterns.

6.2.1 Overall false-positive patterns

Expanded Welch improves false-positive rates in selected settings where Normal Wald rejects too often. In others, it further lowers already conservative rates. Table 6.2 summarises the 120 settings at common MI 0.02 nats.

We classify Expanded Welch as improving nominal calibration at a setting when its false-positive rate is closer to 5% than Wald's by more than 0.3 percentage points. It worsens nominal calibration when its rate is farther from 5% by more than 0.3 percentage points. Differences within this tolerance are grouped separately. All 29 improvements occur with Wald above 5%; 48 of the 52 deteriorations occur with Wald below 5%.

Table 6.2: Change in distance from 5% rejection across the broad null grid.

Expanded Welch relative to Normal Wald	Settings
Closer to 5%	29
Farther from 5%	52
Change within the tolerance	39

Moving farther from 5% is more common than moving closer across this grid. The larger cutoff therefore often intensifies conservatism, even though it reduces excessive rejection at other settings.

Reported rejection rates divide valid rejections by all simulation repetitions, counting invalid outputs as non-rejections. Brackets around rates and shading in rejection plots give pointwise 95% intervals for uncertainty from the finite simulation batch. Figures 6.4 and 6.5 show uniform and different-skew margins at common MI 0.02 nats. Each table-size panel plots false-positive rates against sample size, with a black line marking the 5% target. Vertical scales differ between panels, so comparisons between table sizes should use the labelled rates. The lower strips show the fraction of valid outputs. Because invalid results count as non-rejections, these strips help identify rates that also reflect failed calculations.

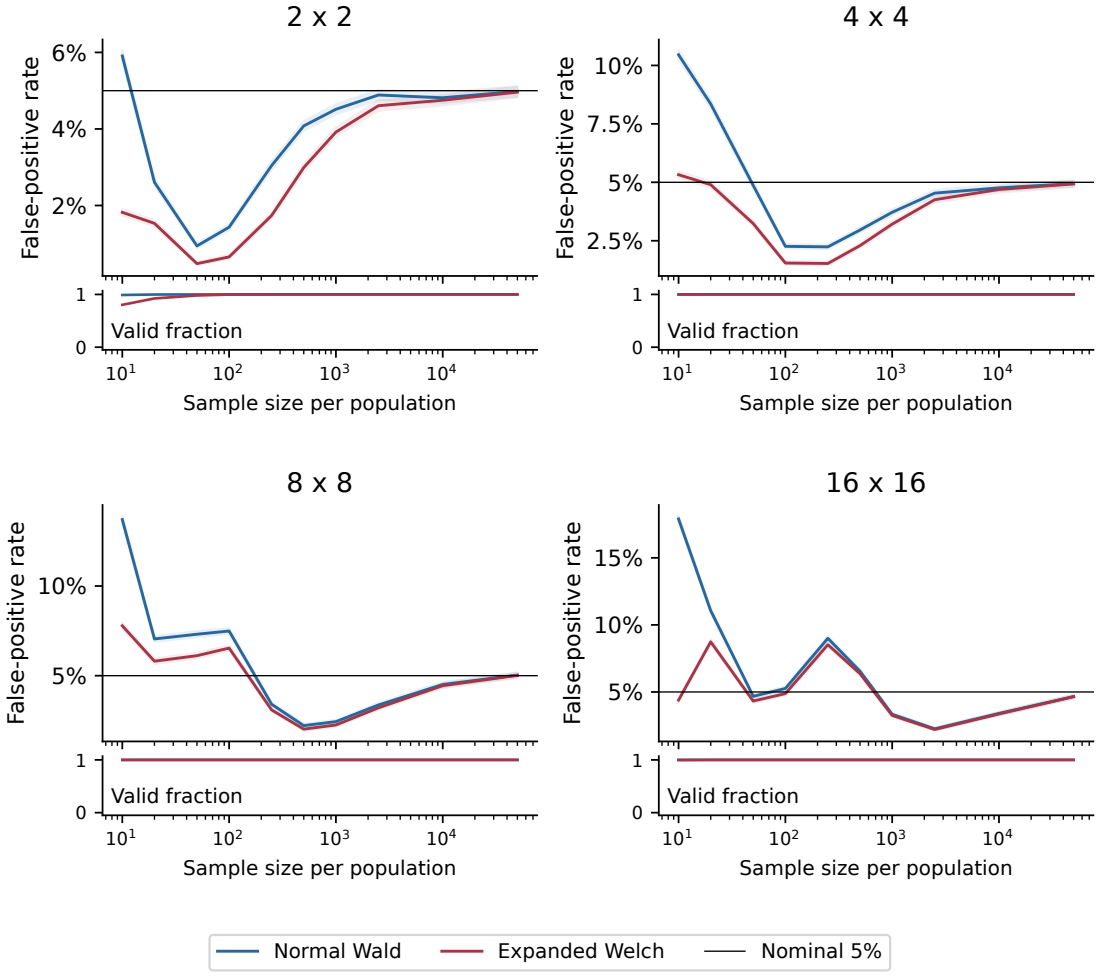


Figure 6.4: False-positive rates under uniform margins.

In Figure 6.4, uniform margins give $P = Q$, so the expected MI estimation biases cancel in the difference at every plotted sample size. The four panels therefore show how testing can fail even without difference bias. Errors in the shared standard error and the tails of T are examined in Section 6.2.2.

In the 2×2 panel, both rates fall into a deep trough before recovering towards 5%. Expanded Welch is more conservative throughout most of this range. The 4×4 panel begins with excessive Wald rejection, which Expanded Welch brings close to the target. Both then become conservative, and the same larger cutoff moves Expanded Welch farther below 5%. The benefit of lowering rejection thus changes within one panel as sample size increases.

The 8×8 panel has an initial shoulder of excessive rejection, followed by a later conservative trough and recovery. Expanded Welch reduces the initial excess but deepens the trough. The 16×16 panel has a more irregular progression: early excess falls to around 5%, rises again, and then gives way to conservatism before recovering. These crossings accompany changes in whether the estimated standard error understates or overstates sampling variation. Increasing sample size does not make those errors shrink monotonically. Across the larger-table panels, the methods draw close well before both rates settle near 5%, so agreement between tests does not establish good calibration.

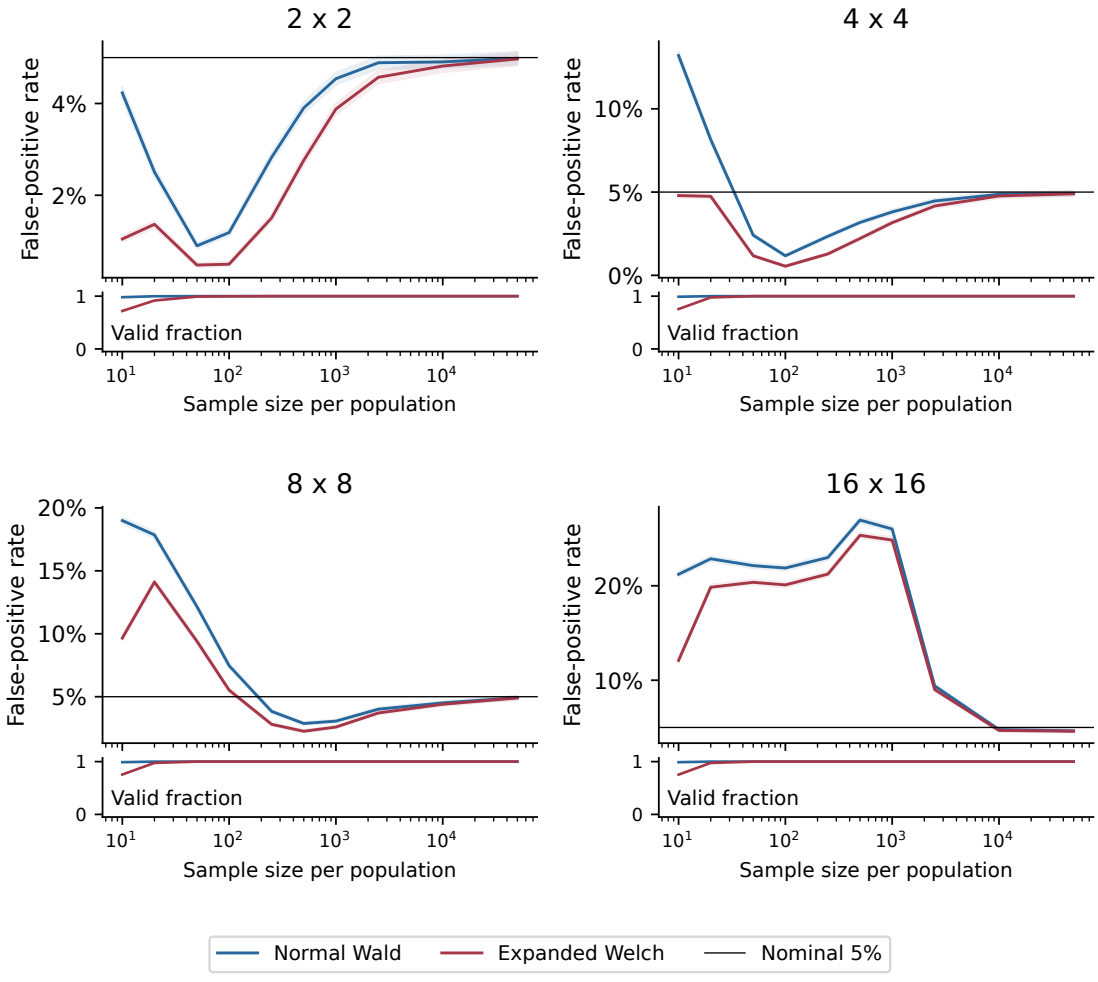


Figure 6.5: False-positive rates under different-skew margins.

In Figure 6.5, the two smaller-table panels also have a trough followed by recovery. In 2×2, even Wald remains below 5% throughout the grid, so Expanded Welch mainly increases conservatism. In 4×4, Wald starts above the target and then falls below it; Expanded Welch lowers both regions. These shapes resemble their uniform counterparts despite the different population tables.

The 8×8 panel starts with much greater excessive rejection than under uniform margins. Both methods eventually fall below 5% and then recover, but Expanded Welch remains visibly more conservative through the intermediate range. The bias diagnostic in the next subsection shows that its early improvement at B accompanies both a shifted difference estimate and an overestimated squared standard error. Lowering rejection alone does not identify which error is being offset.

In contrast, the 16×16 panel shows severe over-rejection across several intermediate sample sizes. The two populations now have different cell probabilities, so their MI estimation biases need not cancel. The bias in the difference grows relative to its sampling SD over much of this range, shifting T away from zero. Expanded Welch's larger cutoff reduces rejection but leaves this shift unchanged. The bias later diminishes and rejection falls towards 5%. Section 6.2.2 gives the supporting diagnostics.

At the smallest sample sizes in Figure 6.5, closeness to 5% can also accompany a loss of usable results. For different-skew 4×4 tables at $n = 10$, Expanded Welch's rate is close to 5%, but it produces valid outputs in only about three quarters of repetitions. Across the full 120-setting grid, the minimum valid fractions are 0.958 for Normal Wald and 0.631 for Expanded Welch. The supplement reports rejection rates conditional on validity and the included counts separately.

The validity strips in Figures 6.4 and 6.5 distinguish these early failures from the later patterns. Under uniform margins, appreciable losses are concentrated in the smallest 2×2 samples; the other strips stay near one. Under different skew, several table sizes lose outputs at the smallest sample sizes. The conservative troughs in 4×4 and larger tables, and the different-skew 16×16 plateau, persist after validity is complete. Failed calculations therefore cannot account for those parts of the curves.

At the largest sample size in both figures, both tests' rates are close to 5% across all table-size panels. Better-estimated cell frequencies reduce the distortions of the shared statistic, while larger estimated degrees of freedom bring Expanded Welch's cutoff closer to Wald's. The curves therefore draw together as well as approaching the target in these settings.

6.2.2 Bias and standard-error accuracy

RQ1 checked the moments of $\widehat{V}(P)$ and $\widehat{V}(Q)$ used to choose degrees of freedom. We now check the shared estimates entering T: the centre of $\widehat{\Delta}_{\mathrm{BC}}$ and whether $\widehat{\mathrm{SE}}^2$ describes its observed sampling variance. Because $V(P)/n_P$ is itself a first-order approximation, an upward error in $\widehat{V}(P)$ relative to $V(P)$ can coexist with an underestimated standard error for the MI difference.

Difference bias measures how far the average $\widehat{\Delta}_{\mathrm{BC}}$ lies from Δ. The variance ratio for $\widehat{\Delta}_{\mathrm{BC}}$ divides the average $\widehat{\mathrm{SE}}^2$ by the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$. A ratio below one means that the squared standard error underestimates sampling variance; a ratio above one means that it overestimates sampling variance. These diagnostics measure difference bias and average error in the squared standard error, both of which can affect rejection. Table 6.3 compares the focus settings and the sparse 16×16 cases at $n = 1{,}000$. The full diagnostics, including the larger-sample 16×16 cases, remain in Table C.3.

Table 6.3: Bias and standard-error accuracy at selected equal-MI settings.

Setting	Bias / SD	$\mathrm{E}(\widehat{\mathrm{SE}}^2)/\,\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$
A	0.003	0.827
B	0.870	1.535
C	0.000	1.010
16×16, same skew, $n = 1{,}000$	-0.003	1.929
16×16, different skew, $n = 1{,}000$	2.058	1.938

Expanded Welch brings rejection close to 5% at A. At B it also reduces rejection, although the rate remains above 5%. Table 6.4 compares these settings with the larger-sample and sparse-table cases. Both tests are valid in all 100,000 repetitions at each of these seven settings.

Table 6.4: Null rejection rates with pointwise 95% intervals.

Setting	Normal Wald	Expanded Welch
A	8.35% [8.18, 8.53]	4.90% [4.76, 5.03]
B	7.47% [7.30, 7.63]	5.52% [5.38, 5.67]
C	4.87% [4.73, 5.00]	4.77% [4.64, 4.90]
16×16, same skew $n = 1{,}000$	0.60% [0.56, 0.65]	0.53% [0.48, 0.57]
16×16, same skew $n = 10{,}000$	4.38% [4.26, 4.51]	4.26% [4.14, 4.39]
16×16, different skew $n = 1{,}000$	26.01% [25.74, 26.28]	24.84% [24.57, 25.10]
16×16, different skew $n = 10{,}000$	4.78% [4.65, 4.91]	4.68% [4.55, 4.82]

In Table 6.3, A's difference bias is negligible, but the average $\widehat{\mathrm{SE}}^2$ is below the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$. At B, $\widehat{\Delta}_{\mathrm{BC}}$ is biased upwards and the average $\widehat{\mathrm{SE}}^2$ exceeds the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$. Expanded Welch lowers false positives in both cases despite these different estimation errors. At C, bias is negligible and the variance ratio is close to one. Table C.3 gives the full diagnostics and degrees-of-freedom comparisons. Returning to Figure 6.4, the uniform-table curves help separate errors in $\widehat{\mathrm{SE}}^2$ from difference bias. Across their sample-size grid, the difference remains centred near zero, but the average $\widehat{\mathrm{SE}}^2$ can move from below to above the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$. For a given estimated difference, a smaller standard error increases the magnitude of T. Underestimating the sampling variance therefore helps explain excessive rejection, while overestimating it can produce conservatism. In the 16×16 panel, the repeated crossings accompany changes in the direction of this variance error. The estimates entering the statistic recover at different rates as the sparse sampled tables fill, so its rejection rate need not improve monotonically.

In that figure's 2×2 trough, average variance accuracy does not fully determine the tails of T. Sampled datasets with larger absolute MI differences tend also to have larger standard errors. Dividing by those larger standard errors reduces the extreme values of T, helping explain why rejection can be far below 5% even when the average $\widehat{\mathrm{SE}}^2$ is fairly close to the observed difference variance. Beyond the trough, this dependence weakens and the tails move closer to the normal approximation. Appendix C.1 reports the diagnostic checks. These checks explain the broad patterns without isolating every finite-sample contribution to rejection.

Both tests also over-reject at stronger common MI. Table 6.5 retains A's and B's margins, table sizes and sample sizes while increasing the common MI. Rates and their pointwise 95% intervals are percentages of all 100,000 repetitions.

Table 6.5: False-positive rates at stronger common MI.

Setting	Common MI (nats)	Normal Wald	Expanded Welch
A	0.3299	16.16 [15.94, 16.39]	13.56 [13.35, 13.78]
B	0.2806	7.62 [7.46, 7.79]	7.25 [7.09, 7.41]

The two rows show different sensitivity to the common MI. At A, the stronger association raises both rates well above their values at MI 0.02, and Expanded Welch no longer brings rejection close to 5%. At B, Wald's rate changes little, but Expanded Welch's earlier reduction becomes much smaller; both remain above the target. The larger cutoff still reduces rejection in each row, but the improvement at common MI 0.02 does not extend to reliable calibration at these stronger MI values.

The sparse different-skew rows of Tables 6.3 and 6.4 show that changing the degrees of freedom does not resolve the failure. At 16×16, different skew and $n = 1{,}000$, the difference bias is about 0.030 nats, or 2.06 observed SDs. Both tests reject far too often; changing the rejection cutoff leaves the bias unchanged. The sampling centre is $\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) = \Delta + \mathrm{Bias}(\widehat{\Delta}_{\mathrm{BC}})$. Under the equal-MI null, this gives

$$\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) \approx 0 + 0.02985 = 0.02985 \text{ nats}. \tag{6.1}$$

This shift is large relative to the observed SD of approximately 0.01450 nats. It moves most rejections into the positive tail even though the population difference is zero.

The sparse same-skew and different-skew rows have almost the same variance ratio in Table 6.3, yet their rejection rates lie on opposite sides of 5%. The distinction is the centre of the estimated difference. Under equal MI, the same-skew populations have identical tables. With equal sample sizes, their expected MI estimation biases therefore cancel in the difference. The average $\widehat{\mathrm{SE}}^2$ exceeds the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, and both rejection rates are far below 5%. Large difference bias accompanies excessive rejection in the different-skew case; in the same-skew case, that bias cancels and both tests are conservative. At $n = 10{,}000$, both skew regimes have much smaller estimation errors and rates closer to 5%.

The sparse-table failure also occurs under the four-cell construction. At 16×16, common MI 0.02 nats and $n_P = n_Q = 1{,}000$, replacing copying by a probability

transfer within four cells preserves each population's margins and MI. Under different-skew margins, the estimated difference remains substantially biased and both false-positive rates are near 30%. The experimental supplement reports the full construction comparison.

6.3 RQ2.2: Nominal and calibrated power

The power comparison uses A (uniform 4×4, $n_P = n_Q = 20$), B (different-skew 8×8, $n_P = n_Q = 100$) and C (different-skew 4×4, $n_P = n_Q = 10{,}000$). For each setting, σ_0 is the independently fitted SD of $\widehat{\Delta}_{\mathrm{BC}}$ under $I(P) = I(Q) = 0.02$ nats. We hold $I(P) = 0.02$ and increase $I(Q)$, so $\Delta = I(P) - I(Q)$ is negative under the alternatives. The horizontal axis is $-\Delta/\sigma_0$: moving right increases the MI advantage of Q in fitted null SD units. Nominal power uses each test's usual 5% cutoff; calibrated power uses cutoffs fitted to target 5% rejection at a specified equal-MI setting.

6.3.1 Nominal power

Expanded Welch has lower nominal power than Normal Wald at the selected small-sample settings A and B (Table 5.3). Figure 6.6 shows all three focus settings with MI differences measured in fitted null standard deviations. The nominal comparison includes their different null rejection rates at zero. Appendix C.2 gives numerical rates and differences between methods evaluated on the same simulated data.

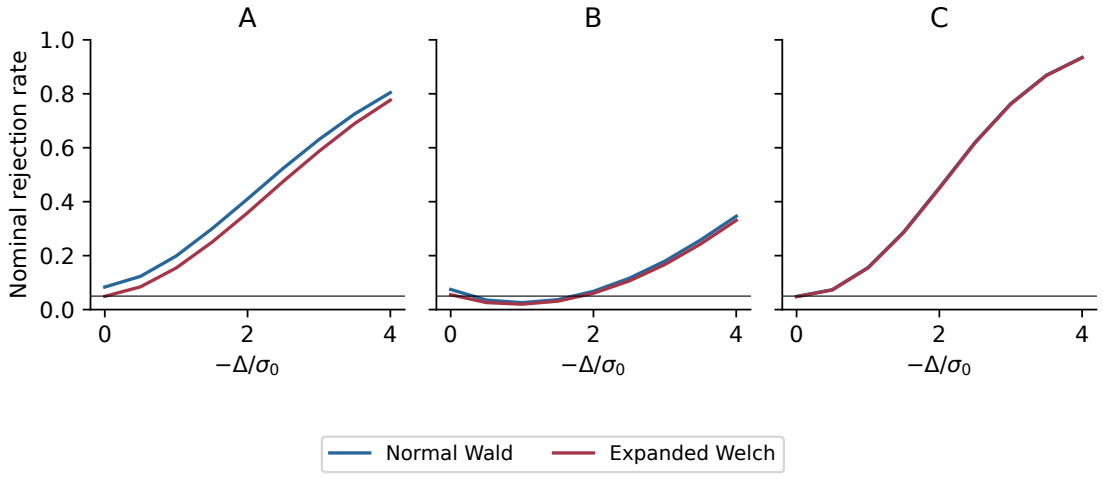


Figure 6.6: Nominal power across differences measured in fitted null standard deviations.

In panels A and C of Figure 6.6, increasing $I(Q)$ moves the sampling centre of $\widehat{\Delta}_{\mathrm{BC}}$ farther below zero. Its magnitude grows relative to the standard error, so

more repetitions cross the negative rejection boundary and power rises. Expanded Welch's larger cutoff leaves some of these repetitions unrejected, placing its curve below Wald's.

At A, the separation between methods is largest over the middle of the curve. More repetitions then lie between the two rejection boundaries. At stronger alternatives, more cross both boundaries, while the sampled degrees of freedom increase and the cutoff from the Student t distribution falls. The power gap consequently narrows again.

As the population MI values move farther apart at B, rejection initially falls before rising. Positive estimation bias persists at the first alternatives. As $I(Q)$ increases, the negative true difference partly offsets this bias, moving the sampling mean of $\widehat{\Delta}_{\mathrm{BC}}$ towards zero. This movement reduces positive-tail rejections before the negative difference becomes large enough to produce many negative-tail rejections. This change in rejection direction helps explain the dip before power rises. Increasing the cutoff leaves this distortion of the shared statistic unchanged. Appendix C reports the estimated directions among rejections in the sparse-table comparisons.

B also remains the least powerful setting farther along the curve. The positive estimation bias continues to offset part of the negative population difference, while the observed SD of $\widehat{\Delta}_{\mathrm{BC}}$ grows relative to its null value. Both reduce the separation of the sampled differences from zero relative to their sampling variation. At C, bias remains negligible and the sampling SD changes less along the alternative path, so power eventually exceeds A's as well as B's. The shared horizontal unit σ_0 standardises the null variation; it does not make bias or alternative sampling variation the same across the three panels.

At C, the estimated degrees of freedom are large enough that the cutoff from the Student t distribution is close to the standard normal cutoff. The methods therefore make nearly the same decisions, and their power curves almost overlap. Table C.5 quantifies the small loss and reports comparisons at two null SDs for all three focus settings. Each interval uses both methods' decisions from the same simulation repetitions to estimate their power difference. These are paired comparisons: the pairing is between methods applied to the same data. The calibrated comparison then assesses power after matching null rejection.

6.3.2 Power after matching false-positive rates

At settings A and B, Expanded Welch's lower nominal power accompanies a lower false-positive rate than Normal Wald's. We now ask whether its power remains lower after adjusting both tests to have approximately the same false-positive rate.

Empirical calibration fits a separate p-value threshold for each test using independent equal-MI simulations to target 5% rejection. These replace the usual threshold of 0.05. We hold them fixed, check their false-positive rates on independent equal-MI samples, and apply them to the same alternative samples used in the nominal comparison. The MI difference $\widehat{\Delta}_{\mathrm{BC}}$, standard error $\widehat{\mathrm{SE}}$ and calculated p-values remain unchanged.

The earlier nesting of rejections applied at a common p-value threshold. With separately fitted thresholds, differences in how the tests rank sampled datasets can give Expanded Welch greater calibrated power.

We first fit the thresholds at $I(P) = I(Q) = 0.02$ nats. On independent null evaluation samples, both tests reject between 4.93% and 4.99% across A–C, so their false-positive rates are closely matched there. Figure 6.7 shows rejection under these thresholds: at zero on the horizontal axis it measures false positives; to the right, it measures power against increasing MI differences.

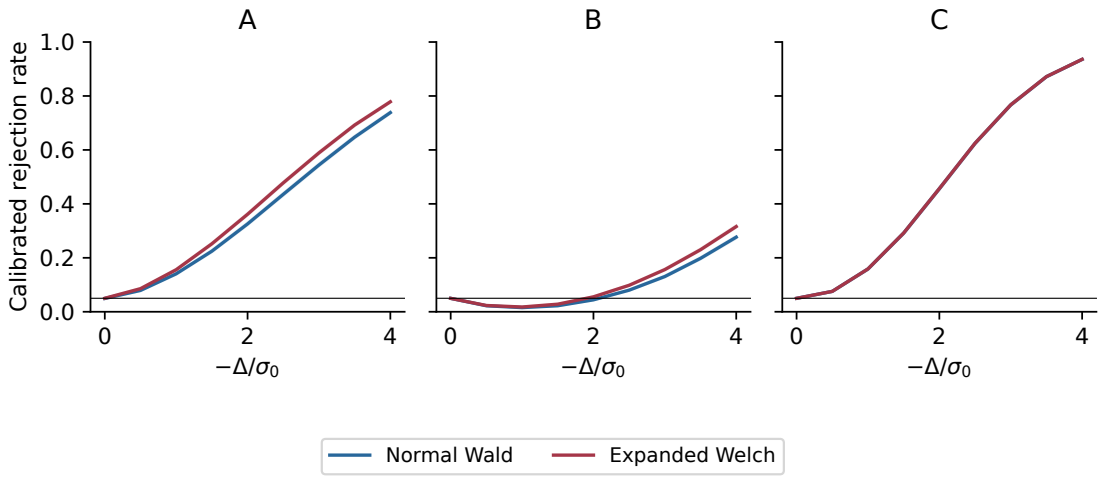


Figure 6.7: Power after calibration at common MI 0.02; bands show bootstrap uncertainty.

In panel A of Figure 6.7, calibration mainly lowers Wald's curve: its nominal false-positive rate was above 5%, whereas Expanded Welch's was already close to the target. Expanded Welch now has higher power over the alternatives shown, reversing the nominal ordering. Its advantage grows over the middle of the curve and narrows slightly at the largest difference.

At B, both curves move down because both nominal false-positive rates were above 5%. Expanded Welch now has greater power over the positive horizontal-axis values, with a larger advantage at the stronger alternatives. Nevertheless, both curves retain the initial dip. Calibration leaves the biased $\widehat{\Delta}_{\mathrm{BC}}$ unchanged, so reversing the method ordering does not remove the distortion in the shared statistic.

At C, calibration makes little change because both nominal false-positive rates were already close to 5%. The curves continue to overlap and rise steeply towards high power. The cross-panel ordering at stronger alternatives also remains: C is most powerful and B least powerful, consistent with their different bias and sampling-variation patterns.

Does the calibration setting change the comparison?

The first comparison matches false-positive rates at common MI 0.02. We next ask whether Expanded Welch's power advantage persists when the thresholds are fitted at a different common MI.

For an alternative with $I(P) = 0.02$ and $I(Q) > 0.02$, we compare:

- **Lower-null calibration:** fit both tests at $I(P) = I(Q) = 0.02$, the alternative's smaller MI.
- **Upper-null calibration:** fit both tests with their population MI values equal to the alternative's larger $I(Q)$.

Both calibration settings retain the alternative's margins and sample sizes. Applying both sets of thresholds to the same alternative samples ensures that only the thresholds change the power comparison.

Table 6.6 reports Expanded Welch power minus Normal Wald power at two and four null SDs. Gains are in percentage points: positive values favour Expanded Welch and negative values favour Wald. The upper-MI column gives the common MI used for upper-null fitting. Brackets give 95% bootstrap intervals accounting for the shared evaluation data and uncertainty from fitting thresholds.

Table 6.6: Expanded Welch minus Wald power under two null calibrations.

Setting, h	Upper-null MI (nats)	Lower-null gain (pp)	Upper-null gain (pp)
A: 2	0.3299	+3.58 [+3.32, +3.80]	-0.76 [-0.91, -0.56]
A: 4	0.6399	+4.01 [+3.79, +4.20]	+0.18 [-0.12, +0.43]
B: 2	0.1503	+1.11 [+1.03, +1.19]	+0.05 [+0.01, +0.10]
B: 4	0.2806	+3.92 [+3.77, +4.14]	-0.39 [-0.48, -0.28]
C: 2	0.0263	+0.04 [+0.01, +0.07]	-0.06 [-0.06, -0.01]
C: 4	0.0327	+0.02 [+0.01, +0.03]	-0.01 [-0.03, -0.01]

In Table 6.6, the lower-null gains at A and B increase from two to four null SDs. Changing the fitting setting nevertheless reverses the gain in two comparisons. At A and two null SDs, the 3.58-point gain becomes a 0.76-point loss. At B and four

SDs, a 3.92-point gain becomes a 0.39-point loss. Under upper-null calibration, A at four SDs retains a small positive estimate whose interval includes zero, while B at two SDs retains a much smaller gain. At C, both calibrations give differences well within the one-percentage-point tolerance.

The calibrated gains therefore depend on where we match false-positive rates. Appendix C.2 reports the rates from applying each calibration to both null settings and the full power tables. Matching at one equal-MI setting does not ensure matching elsewhere. These two checks also do not establish that either test keeps its false-positive rate at or below 5% at every equal-MI setting.

6.4 Application to online shopping data

The Online Shoppers Purchasing Intention dataset records website sessions and whether each session ends in a purchase (Sakar and Kastro, 2018). We use it to address two application questions:

1. **Month:** Does month provide the same amount of information about purchasing among new and returning visitors?
2. **Weekend:** Does weekend status provide the same amount of information about purchasing among new and returning visitors?

The dataset contains 12,330 sessions, with each session belonging to a different user according to the source documentation (Sakar and Kastro, 2018). The analysis retains all 1,694 sessions labelled `New_Visitor` and 10,551 labelled `Returning_Visitor`, excluding the 85 `Other` sessions. The outcome is the original Boolean `Revenue` variable: true indicates a purchase and false indicates no purchase. The tests treat the two disjoint visitor groups as independent samples and the sessions within each group as independent observations.

For each question, X is the session attribute and Y is the purchase outcome. The two populations are defined by visitor type:

Table 6.7: Variables and populations for the two application questions.

	Month question	Weekend question
X	Month of the session	Weekday or weekend
Y	Purchase outcome: no purchase or purchase	
P	Joint distribution of (X, Y) among new visitors	
Q	Joint distribution of (X, Y) among returning visitors	
H_0	$I(P) = I(Q)$ for each choice of X	

For month, each group forms a 10×2 count table. The rows are the ten recorded months: February, March, May, June, July, August, September, October, November and December. The columns are no purchase and purchase. Both groups use these categories, with no smoothing or removal of zero cells. All ten months occur in both groups, although the new-visitor February row contains only one session and no purchase. Replacing month with the Boolean `Weekend` variable gives a 2×2 table for each group, with weekday and weekend rows; all four cells are observed.

The leading MI bias correction subtracts $(10-1)(2-1)/(2n) = 9/(2n)$ from each month plug-in estimate (Miller, 1955; Moddemeijer, 1989), and $1/(2n)$ from each weekend estimate, where n is the relevant group size. The corrected difference $\widehat{\Delta}_{\mathrm{BC}}$ is new-visitor MI minus returning-visitor MI. Table 6.8 gives these differences alongside their shared standard errors, $\widehat{\mathrm{SE}}$, and the resulting $T = \widehat{\Delta}_{\mathrm{BC}}/\widehat{\mathrm{SE}}$. MI estimates, differences and standard errors are in nats.

Table 6.8: Corrected MI estimates and uncertainty of their differences.

	Corrected MI				
Predictor	New	Returning	$\widehat{\Delta}_{\mathrm{BC}}$	$\widehat{\mathrm{SE}}$	T
Month	0.004316	0.017880	-0.013563	0.003333	-4.069
Weekend	0.000664	0.000727	-0.000063	0.001124	-0.056

Month has higher estimated MI with purchasing among returning visitors. The difference lies about 4.07 standard errors below zero:

$$T \approx \frac{-0.013563}{0.003333} \approx -4.069. \tag{6.2}$$

Normal Wald uses the standard normal distribution; Expanded Welch uses a Student t distribution with estimated degrees of freedom. Table 6.9 gives their two-sided p-values, using their usual p-value threshold of 0.05.

Table 6.9: Two-sample test results for online shopping sessions.

		p-value	
Predictor	Expanded Welch degrees of freedom	Normal Wald	Expanded Welch
Month	26.79	4.71×10^{-5}	3.73×10^{-4}
Weekend	2.18	0.9554	0.9601

Both tests reject equal month–purchase MI. Expanded Welch's larger p-value reflects the heavier tails of the Student t distribution with 26.79 degrees of freedom.

For weekend, both corrected MI estimates are approximately 0.0007 nats. Their difference is only about 0.056 standard errors from zero, so both p-values are near one. Neither test provides evidence of different weekend–purchase MI between the groups; this does not establish equality.

Expanded Welch's 2.18 degrees of freedom for weekend reflect substantial estimated sampling uncertainty relative to the sizes of $\widehat{V}(P)$ and $\widehat{V}(Q)$, even with large samples (Figure 6.3). The unknown population tables prevent a direct check of the sampling-moment target.

The shopping comparisons were selected from six exploratory comparisons across three public datasets. Applying Holm's adjustment separately to each method's six p-values (Holm, 1979) gives month p-values of 0.000189 for Normal Wald and 0.00149 for Expanded Welch; both still reject equal month–purchase MI. Table C.9 reports all raw and adjusted p-values.

Chapter 7

Conclusions

7.1 Contributions and main findings

This thesis derives an Expanded Welch–Satterthwaite test for comparing categorical MI across independent populations. Its mathematical contribution is the observation-level sensitivity of the estimated MI variance, accounting for changes in cell probabilities, margins and pointwise MI. The variance of this sensitivity gives the first-order approximation $\mathrm{Var}\{\widehat{V}(P)\} \approx \tau^2(P)/n_P$. Moment matching uses it to choose MI-specific degrees of freedom for a Student t distribution.

7.1.1 RQ1: Accuracy of variance approximations

The predicted sampling variance of $\widehat{V}(P)$ can be close to its observed value even when the sampling mean $\mathrm{E}\{\widehat{V}(P)\}$ remains above $V(P)$. In the moment target in Equation (5.5), the mean is squared in the numerator. Replacing that sampling mean by the smaller population value therefore lowers $\nu_{P,\mathrm{FO}}$ relative to $\nu_{P,\mathrm{mom}}$, even when the sampling-variance approximation is accurate. The sampling mean $\mathrm{E}(\widehat{\nu}_P)$ can also fall below the moment target. The size of these discrepancies depends on the population and sample size.

Near independence, the first-order prediction $\tau^2(P)/n_P$ for $\mathrm{Var}\{\widehat{V}(P)\}$ is small and quadratic sampling terms remain substantial. The calculation in Appendix A.2 explains the observed scale of the estimated component degrees of freedom. These results explain why the first-order approximation becomes inadequate near independence and why substituting sample probabilities requires its own evaluation, even when the population approximation is accurate.

7.1.2 RQ2.1: False-positive rates under equal MI

Expanded Welch brings false-positive rates closer to 5% in the selected small-sample settings A and B, both at common MI 0.02 nats (Table 5.3). Both tests return valid results in every repetition at these settings, so the lower rejection rates reflect the larger cutoff from the Student t distribution. Across the broad grid, Expanded Welch also lowers rates that are already below 5%. Its larger cutoff helps when it corrects excessive rejection and increases conservatism when rejection is already too infrequent.

A false-positive rate near 5% does not establish that the estimated component degrees of freedom match their sampling-moment targets. At A, the average $\widehat{\mathrm{SE}}^2$ is below the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$, while the low estimated degrees of freedom increase the rejection cutoff. These errors can compensate in the false-positive rate. At B, the average squared standard error exceeds the observed variance of the MI difference and the difference is biased, so the same explanation does not apply unchanged.

The sparse same-skew null illustrates how error in the standard error can remain when the expected difference bias cancels. Identical population tables at equal sample sizes have the same expected MI estimation bias. However, the average $\widehat{\mathrm{SE}}^2$ is almost twice the observed $\mathrm{Var}(\widehat{\Delta}_{\mathrm{BC}})$ in the sparse copying case, and both tests are conservative. Unequal sample sizes can also leave unequal residual biases in otherwise identical populations. Reliable testing therefore requires accurate estimation of both the MI difference and its uncertainty.

7.1.3 RQ2.2: Nominal and calibrated power

Expanded Welch has lower nominal power than Normal Wald at the selected small-sample settings A and B. Because the two tests also have different false-positive rates, the calibrated comparison adjusts their cutoffs to match rejection at a specified equal-MI setting before comparing power. After calibration at common MI 0.02 nats, Expanded Welch has higher power at the reported two- and four-null-SD alternatives for A and B. Moving calibration to the upper equal-MI setting while holding the alternative fixed reverses the gain at A and two null SDs, and at B and four null SDs (Table 6.6). At C, nominal power differences are small and calibrated differences remain practically small under both calibrations.

The calibrated power advantage depends on the setting used to fit the cutoffs. Matching false-positive rates at the lower common MI does not ensure that the two tests have the same false-positive rate at the upper common MI. Expanded Welch's

sample-dependent degrees of freedom can change the ordering of datasets by their p-values, but the observed gains do not establish a stable power advantage across the equal-MI null.

Changing the cutoff also leaves errors in the estimated MI difference unchanged. At B, positive estimation bias persists at the first alternatives as the true difference becomes negative. Their partial cancellation moves the sampling mean of $\widehat{\Delta}_{\mathrm{BC}}$ towards zero, so rejection initially falls even though the population MI values move farther apart. This distortion remains after calibration.

The sparse 16×16 different-skew comparison shows a more severe failure. Its positive difference bias exceeds the true negative difference at the 0.01-nat alternative, so almost every rejection gives the wrong estimated ordering. Equation (C.1) explains the sign reversal. The four-cell construction also reproduces the sparse-table failure. A larger rejection cutoff cannot correct that bias.

7.2 Limitations

The derivation assumes independent categorical observations, fixed categories, positive population cell probabilities, and positive $V(P)$, $V(Q)$, $\tau^2(P)$ and $\tau^2(Q)$. Paired samples would require a covariance term in the difference variance. Growing alphabets and continuous MI estimators have different approximation errors. Exact independence requires the second-order argument in Appendix A.2.

The findings concern the tested populations, association constructions and alternative paths. The power settings were selected using earlier false-positive results and evaluated on independent samples. Each curve keeps $I(P) = 0.02$ and increases $I(Q)$ along the copying family. Both tested constructions use equal row and column margins within each table and align association locations between populations. Other directions of MI change, unequal row and column margins within a table and other locations of dependence remain untested. The results therefore support further investigation of Expanded Welch rather than its use as a general replacement for Normal Wald.

Empirical calibration uses known simulation populations. The two calibration checks do not bound rejection across the entire equal-MI null or supply a calibration procedure for arbitrary observed tables. The pointwise confidence intervals quantify simulation uncertainty for the specified populations; the calibrated power intervals also include uncertainty from fitting the cutoffs.

The online-shopping application illustrates how to report the estimated MI difference and its direction, standard error, p-value and Expanded Welch degrees of freedom.

Its unknown population MI values prevent a direct check of false-positive rates. The application therefore demonstrates interpretation of the test outputs, while the simulation evidence establishes their behaviour under known populations.

7.3 Future work

A first extension is to derive the second-order mean contribution to the variance estimator and evaluate a corresponding degrees-of-freedom correction. The observed variation across MI and margins indicates that the correction should depend on the population table. A fixed coefficient based only on table size would leave errors in some settings. Any estimated correction should be evaluated through the resulting test's false-positive rates and power.

Improved MI bias and standard-error estimators could address the sparse-table failures. Their effects should be assessed separately by changing one estimate while holding the others fixed, then measuring null rejection, power and estimated direction. Near independence, the quadratic sampling approximation may support a more accurate comparison of MI differences. Any likelihood-ratio or resampling procedure must preserve the equal-MI null even when the complete population tables differ.

Reversed power paths, stronger dependence at small sample sizes, additional association constructions, unequal margins and unequal sample sizes would assess whether power rankings persist across a wider range of populations. Simulations from plausible application populations could also examine sensitivity to rare cells and distinct equal-MI tables, with conclusions tied to the populations chosen for that assessment.

Appendix A

Near-Independence Calculations

These calculations support the near-independence results in Chapter 4 and their interpretation in Chapter 6. The first calculation varies population association while keeping the margins fixed. The second retains the quadratic sampling variation needed at independence and very small MI.

A.1 Fixed-margin behaviour near independence

Let $P_0 = (q_{ij})$ be independent, with positive cells $q_{ij} = a_i b_j$. Consider $P_t = P_0 + tH$, where every row and column sum of H is zero. For sufficiently small t, all cells remain positive and the margins remain a and b. We expand around $t = 0$, where $I(P_0) = 0$.

For $f(t) = I(P_t)$, the second-order Taylor formula is

$$f(t) = f(0) + f'(0)t + \frac{f''(0)}{2!}t^2 + O(t^3). \tag{A.1}$$

Here $O(t^3)$ collects the omitted terms whose magnitude is bounded by a constant times $|t|^3$ as t approaches zero. These terms become smaller than the retained quadratic term. The same notation below describes the size of each omitted remainder near zero.

The fixed margins make the pointwise MI in cell (i, j)

$$\ell_{P_t}(i,j) = \log\left(\frac{q_{ij} + tH_{ij}}{q_{ij}}\right) = \log\left(1 + \frac{tH_{ij}}{q_{ij}}\right) = \frac{tH_{ij}}{q_{ij}} - \frac{t^2 H_{ij}^2}{2q_{ij}^2} + O(t^3), \tag{A.2}$$

using $\log(1+z) = z - z^2/2 + O(z^3)$ with $z = tH_{ij}/q_{ij}$. Multiplying by the cell probability gives

$$\begin{aligned}(q_{ij} + tH_{ij})\ell_{P_t}(i,j) &= (q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} - \frac{t^2H_{ij}^2}{2q_{ij}^2} + O(t^3)\right\} \\ &= tH_{ij} - \frac{t^2H_{ij}^2}{2q_{ij}} + \frac{t^2H_{ij}^2}{q_{ij}} + O(t^3) \\ &= tH_{ij} + \frac{t^2H_{ij}^2}{2q_{ij}} + O(t^3).\end{aligned}$$

Multiplication by q_{ij} supplies the negative quadratic term; multiplication by tH_{ij} supplies the positive one. Their combination leaves half the positive term. When we sum over cells, the first-order terms cancel because $\sum_{i,j} H_{ij} = 0$. Thus $f(0) = f'(0) = 0$, $f''(0) = \sum_{i,j} H_{ij}^2/q_{ij}$, and

$$I(P_t) = \frac{t^2}{2}\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.3}$$

To calculate $V(P_t)$, first square the pointwise-MI expansion:

$$\ell_{P_t}(i,j)^2 = \left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 = \frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3).$$

Multiplying by the cell probabilities and summing gives its second moment:

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y)^2\} &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{t^2H_{ij}^2}{q_{ij}^2} + O(t^3)\right\} \\ &= t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned}$$

MI itself begins with a quadratic term, so $I(P_t)^2 = O(t^4)$. Subtracting it from this second moment leaves the quadratic term unchanged:

$$V(P_t) = t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3). \tag{A.4}$$

The row conditional mean also has no first-order term. Its denominator remains a_i because the margins are fixed, and

$$\begin{aligned}\mathrm{E}_{P_t}\{\ell_{P_t}(X,Y) \mid X = i\} &= \frac{1}{a_i}\sum_{j}(q_{ij} + tH_{ij})\left\{\frac{tH_{ij}}{q_{ij}} + O(t^2)\right\} \\ &= \frac{t}{a_i}\sum_{j}H_{ij} + O(t^2) = O(t^2).\end{aligned}$$

The last equality uses the zero row sum of H. The column mean has the same cancellation, using $\sum_i H_{ij} = 0$.

In Equation (4.30), the squared centred pointwise MI, $V(P_t)$, the conditional means and $I(P_t)$ all begin at order t^2. Only the unsquared pointwise-MI term in the bracket has a first-order contribution. Therefore Equation (4.30) supplies the first-order term:

$$g_{P_t}(i,j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \tag{A.5}$$

Since g_{P_t} has mean zero, its variance is its second moment:

$$\begin{aligned}\tau^2(P_t) &= \sum_{i,j}(q_{ij} + tH_{ij})\left\{\frac{2tH_{ij}}{q_{ij}} + O(t^2)\right\}^2 \\ &= 4t^2\sum_{i,j}\frac{H_{ij}^2}{q_{ij}} + O(t^3).\end{aligned} \tag{A.6}$$

Comparing the leading terms of $I(P_t)$, $V(P_t)$ and $\tau^2(P_t)$ gives $V(P_t) \approx 2I(P_t)$ and $\tau^2(P_t) \approx 4V(P_t)$ for small positive MI along this path. At $t = 0$, the first-order sampling variance is zero; Appendix A.2 treats that case separately.

A.2 The independence boundary

At independence, $p_{ij} = p_{i+}p_{+j}$, so every population pointwise-MI value is $\ell_P(i,j) = \log 1 = 0$. Consequently, $I(P) = V(P) = 0$, and the first-order change in MI is zero. Sampling can still produce positive estimated MI, but its leading variation comes from second-order cell errors.

To calculate this variation, fix a strictly positive independent table P and let $h = (h_{ij})$ describe a redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define

$$P_\varepsilon = P + \varepsilon h, \qquad f(\varepsilon) = I(P_\varepsilon), \tag{A.7}$$

for ε small enough that every cell remains positive. The expansion point is $\varepsilon = 0$, where $f(0) = I(P) = 0$. The second-order Taylor formula gives

$$f(\varepsilon) = f(0) + f'(0)\varepsilon + \frac{f''(0)}{2!}\varepsilon^2 + o(\varepsilon^2). \tag{A.8}$$

The first derivative is

$$f'(0) = \sum_{i,j}\{\ell_P(i,j) - 1\}h_{ij} = -\sum_{i,j} h_{ij} = 0. \tag{A.9}$$

Here the pointwise-MI terms vanish at independence, and the remaining terms cancel because the probabilities must still sum to one.

The row and column changes are $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Applying $\mathrm{d}^2(u \log u)/\mathrm{d}u^2 = 1/u$ to the joint and marginal terms in Equation (2.11) gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \tag{A.10}$$

Thus the leading change is

$$I(P_\varepsilon) = \frac{\varepsilon^2}{2} \left\{ \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}} \right\} + o(\varepsilon^2). \tag{A.11}$$

The first sum measures squared cell changes, whereas the last two remove the contributions due to changed margins, leaving the change in association. The remainder is negligible relative to ε^2 as ε approaches zero.

For a multinomial sample, take $\varepsilon h = \widehat{P} - P$. Since cell-probability errors typically have size proportional to $n^{-1/2}$, squaring them makes $\widehat{I}(P)$ proportional to n^{-1} under independence. Under fixed table dimensions and positive cell probabilities, the likelihood-ratio result is

$$2n\widehat{I}(P) \xrightarrow{d} \chi^2_{(r-1)(c-1)} \tag{A.12}$$

(Wilks, 1938; Brillinger, 2004). This is the usual one-table likelihood-ratio, or G, test of independence. Its chi-squared distribution comes from quadratic cell errors, whereas Normal Wald and Expanded Welch use the first-order MI error. The two-sample first-order approximation therefore does not apply unchanged at exact independence.

Sampling variation at small positive MI.

The quadratic approximation also explains why estimated degrees of freedom remain several units when population MI is almost zero. Keep r and c fixed and let a positive population table approach independence with cell departures proportional to $n^{-1/2}$. The normal sampling errors and the population departure then contribute on the same scale. In regular models, the likelihood-ratio statistic has a noncentral chi-squared limit under local alternatives (Banerjee, 2005). For the multinomial model here, assume strictly positive limiting cell probabilities. The local approximation is

$$G = 2n\widehat{I}(P) \dot{\sim} \chi^2_d(\kappa), \qquad d = (r-1)(c-1), \qquad \kappa \simeq 2nI(P), \tag{A.13}$$

where $\dot{\sim}$ denotes an approximate distribution. The noncentral chi-squared distribution describes a sum of squared normal errors with a nonzero mean departure; κ measures that departure. At independence, $\kappa = 0$, recovering Equation (A.12).

For a sampled table close to independence, the expansions in Appendix A.1, applied around its own independent margins, give

$$\widehat{V}(P) \approx 2\widehat{I}(P) = \frac{G}{n}, \qquad \widehat{\tau}^2(P) \approx 4\widehat{V}(P) \approx \frac{4G}{n}. \tag{A.14}$$

Substitution into the implemented component formula gives

$$\widehat{\nu}_P \approx \frac{2n(G/n)^2}{4G/n} = \frac{G}{2}. \tag{A.15}$$

Thus sampling noise contributes to the estimated degrees of freedom even when the population first-order approximation $nI(P)$ is very small.

The same approximation separates that plug-in calculation from matching the actual sampling moments. Under the noncentral chi-squared approximation, $\mathrm{E}(G) \approx d + \kappa$ and $\mathrm{Var}(G) \approx 2(d + 2\kappa)$, giving

$$\mathrm{E}\{\widehat{V}(P)\} \approx \frac{d+\kappa}{n}, \qquad \mathrm{Var}\{\widehat{V}(P)\} \approx \frac{2(d+2\kappa)}{n^2}. \tag{A.16}$$

Keeping both terms in the moment-matching equation gives

$$\nu_{P,\mathrm{mom}} \approx \frac{2\{(d+\kappa)/n\}^2}{2(d+2\kappa)/n^2} = \frac{(d+\kappa)^2}{d+2\kappa}. \tag{A.17}$$

When κ is much larger than d, this reduces to $\kappa/2 \simeq nI(P)$, as in Equation (4.52). When κ is small, however, the matched degrees of freedom approach d, while the mean plug-in degrees of freedom approach $d/2$.

For the 4×4, $n = 1{,}000$, $I(P) = 0.0001$ setting, $d = 9$ and $\kappa \simeq 0.2$. The local calculation gives a mean estimated component degree of freedom of approximately 4.6, rather than the population first-order value 0.1. It also predicts that the average plug-in estimate of $\mathrm{Var}\{\widehat{V}(P)\}$ divided by its actual sampling variance is approximately

$$\frac{4(d+\kappa)/n^2}{2(d+2\kappa)/n^2} = \frac{2(d+\kappa)}{d+2\kappa} \simeq 1.96. \tag{A.18}$$

This explains the scale of the estimated degrees of freedom and the roughly twofold plug-in overprediction in Table 6.1. The population formula alone omits the quadratic sampling contribution responsible for both effects.

Appendix B

Experimental Details

This appendix specifies the population construction, target-MI search, settings underlying the main results, sampling procedure and outcome calculations. Together with the test formulas in Chapters 3–5, these details define the statistical experiment.

B.1 Copying construction and target selection

For the copying construction in Equation (5.1), summing a row gives

$$\sum_j p_{ij}(\lambda) = (1-\lambda)a_i \sum_j a_j + \lambda a_i = a_i. \tag{B.1}$$

Summing a column similarly gives a_j, so both margins are preserved. All cells are positive for $0 \leq \lambda < 1$.

MI increases continuously from zero to the entropy of a. With the margins fixed, differentiating the joint terms gives

$$\frac{\mathrm{d}^2 I\{P(\lambda)\}}{\mathrm{d}\lambda^2} = \sum_{i,j} \frac{(a_i \mathbf{1}_{\{i=j\}} - a_i a_j)^2}{p_{ij}(\lambda)} > 0. \tag{B.2}$$

The first derivative at zero is zero, because the change in probabilities sums to zero and all pointwise MI values are zero there. The positive second derivative makes the first derivative positive for $\lambda > 0$. At the endpoint, $Y = X$, so

$$I\{P(1)\} = -\sum_i a_i \log a_i. \tag{B.3}$$

These facts give a unique copying probability for each positive target below the entropy endpoint.

For each population, calculate this endpoint and solve $I\{P(\lambda)\} - I_{\text{target}} = 0$ on $[0, 1]$ using Brent's method (Brent, 1973). Absolute and relative root tolerances are 10^{-15} and 10^{-14}. The endpoint brackets the search; sampled populations use the interior solution. Accept a table only when every cell is positive, its total and marginal errors are at most 10^{-12}, and its MI error is at most $10^{-12} + 10^{-8} I_{\text{target}}$. Every target is checked against its own population entropy endpoint, including the alternatives scaled by the fitted null standard deviation.

B.2 Four-cell supporting construction

For the supporting 4×4 and 16×16 comparisons, begin with the independence probabilities $a_i a_j$ and transfer an amount t within the first two rows and columns. The resulting table $R(t)$ has cell probabilities

$$r_{ij}(t) = \begin{cases} a_i a_j + t, & (i, j) = (1, 1) \text{ or } (2, 2), \\ a_i a_j - t, & (i, j) = (1, 2) \text{ or } (2, 1), \\ a_i a_j, & \text{all other cells.} \end{cases} \tag{B.4}$$

Every affected row and column gains and loses t, preserving its total. The decreasing cells have starting probability $a_1 a_2$, so

$$0 \leq t < t_{\max} = a_1 a_2. \tag{B.5}$$

Solve $I\{R(t)\} = I_{\text{target}}$ by the same root search on $[0, t_{\max}]$. The target must lie below the endpoint MI, and the constructed table passes the same checks as the copying table.

B.3 Parameter grids and sampling

Table B.1 lists the settings underlying the main results. The experimental supplement specifies the broader power and sample-allocation comparisons. Table sizes are $k \in \{2, 4, 8, 16\}$. Each population has identical row and column margins. Uniform margins give probability $1/k$ to every category; same skew uses $m_k(0.8)$ in both populations; different skew uses $m_k(0.7)$ in P and $m_k(0.8)$ in Q, with m_k defined by Equation (5.3). The same copying formula and root search apply to every primary-study setting.

Table B.1: Exact parameter sets; MI values are in nats.

Study	Settings
Equal MI	All four sizes and three margin regimes; $I(P) = I(Q) = 0.02$; equal sample sizes $n \in \{10, 20, 50, 100, 250, 500, 1000, 2500, 10000, 50000\}$.
Population MI	4×4, uniform, same skew and different skew; $I(P) = I(Q) \in \{0.0001, 0.0003, 0.001, 0.003, 0.01, 0.02, 0.05\}$; $n \in \{100, 1000, 10000\}$.
Focused power	Settings A–C in Table 5.3; $I(P) = 0.02$, $I(Q) = 0.02 + h\sigma_0$, $h \in \{0, 0.5, 1, 1.5, 2, 2.5, 3, 3.5, 4\}$. The fitted null SDs defining these populations are given below.
Calibration sensitivity	Settings A–C at $h \in \{2, 4\}$; upper null $I(P) = I(Q) = 0.02 + h\sigma_0$, retaining each setting's margins, sample sizes and copying construction. Both calibrations are applied to the fixed alternatives in the focused-power row.
Four-cell check	$k \in \{4, 16\}$, same skew and different skew; $(I(P), I(Q)) \in \{(0.02, 0.02), (0.02, 0.03)\}$; $n \in \{1000, 10000\}$; construction in Equation (B.4).

For the focused power comparison, σ_0 is the SD of $\widehat{\Delta}_{\mathrm{BC}}$ estimated from the independent null fitting batch, centred at its sampling mean. Equation (5.6) defines the alternatives in multiples of this SD. The fixed null-SD constants used to construct the focused alternatives are

$$\sigma_{0,A} = 0.154966805570126 \text{ nats},$$
$$\sigma_{0,B} = 0.065137776499116 \text{ nats},$$
$$\sigma_{0,C} = 0.003174953145465 \text{ nats}.$$

For each h in the focused grid, set the target MI of Q to $0.02 + h\sigma_0$ and solve for its copying probability. The fitting uncertainty of these constants is reported in Table C.4; the constructed populations remain fixed during evaluation.

For one repetition, draw

$$(N_{P,11}, \ldots, N_{P,kk}) \sim \mathrm{Multinomial}(n_P; p_{11}, \ldots, p_{kk}), \tag{B.6}$$

and independently draw the corresponding Q counts. Reshape these counts into their $k \times k$ tables and apply both tests to those same two sampled tables. Repeat independently 100,000 times per distinct evaluation setting. Generate a separate 100,000-repetition fitting batch for each equal-MI setting in the focused power comparison. The simulations use the PCG64 pseudo-random generator with master seed 20261021 and separate streams for population, setting, fitting and evaluation samples. Repeated settings use one set of repetitions in every relevant display.

All samples retain their declared table dimensions, including empty rows and columns. Use natural logarithms, $0 \log 0 = 0$, and no smoothing. The bias correction for each plug-in MI is $(k-1)^2/(2n)$; do not truncate negative corrected estimates. In the equal-sample studies, the two bias corrections cancel in their difference. The experimental supplement gives the corrections for the unequal-sample comparison. Calculate the standard error using Equation (3.9) and the component and combined degrees of freedom using Chapter 4.

B.4 Validity and sparse-sample measures

Normal Wald requires a finite corrected difference, a finite positive standard error, $\widehat{V}(P)+\widehat{V}(Q) > 10^{-14}$, and a finite p-value. The threshold treats variances extremely close to zero as numerically unusable. Expanded Welch additionally requires positive finite degrees of freedom for both components and their combination, and a finite expanded p-value. A failed component makes Expanded Welch invalid under this rule.

For positive component degrees of freedom, substituting Equation (4.44) and its Q analogue into Equation (4.45) gives the equivalent combined form

$$\widehat{\nu}_{\text{expanded}} = \frac{\left\{\widehat{V}(P)/n_P + \widehat{V}(Q)/n_Q\right\}^2}{\widehat{\tau}^2(P)/(2n_P^3) + \widehat{\tau}^2(Q)/(2n_Q^3)}. \tag{B.7}$$

This direct form remains defined when one table has $\widehat{V} = \widehat{\tau}^2 = 0$ and the other gives a positive denominator. The zero contribution drops out, leaving the other table's component degrees of freedom. For the zero component, however, Equation (4.44) requires $0/0$. The specified componentwise algorithm requires both component degrees of freedom to be positive and finite, so it records this case as invalid even though the direct combined expression remains defined.

The primary rate divides valid rejections by all repetitions. A method-valid rate uses only that method's valid repetitions; a common-valid rate uses only repetitions where both tests are valid. Report the denominator for either conditional rate,

which is undefined when the denominator is zero. Invalid outputs contribute no rejection to the primary rate.

Record the number of zero-count cells, empty rows and empty columns separately for each population. These are sampling diagnostics rather than eligibility rules. The minimum expected cell count is $\min_{ij} np_{ij}$, with the corresponding definition for Q. The study applies no minimum expected-count filter.

B.5 Bias, variance and degrees of freedom

Let $\widehat{\Delta}_s$ be the corrected difference in repetition s. Its bias estimate is the mean of the finite-valued MI difference estimates minus the known Δ. Its observed sampling variance is

$$\frac{1}{M-1}\sum_{s=1}^{M}\left(\widehat{\Delta}_s - \frac{1}{M}\sum_{u=1}^{M}\widehat{\Delta}_u\right)^2, \tag{B.8}$$

where M counts the finite-valued differences, including those whose test output is invalid. Centring at their sample mean separates variance from bias. For P and Q, the individual corrected MI biases are the mean finite corrected MI estimates minus $I(P)$ and $I(Q)$, respectively.

For the estimated / observed variance ratio, restrict the calculations to the same repetitions with a finite difference and finite squared standard error. Retain finite zero variance estimates. Divide the mean squared standard error by the observed variance on this matched subset. Report the included fraction. The ratio is defined only when the observed variance is positive. The population comparison instead uses $V(P)/n_P + V(Q)/n_Q$ and the primary observed variance.

The Monte Carlo standard error of estimated bias is the observed SD divided by $\sqrt{M}$. Component and combined degrees-of-freedom summaries retain positive finite values and report their included fractions. Finite zero variance estimates remain in moment calculations even when component degrees of freedom are undefined. Percentile ranges describe sampling spread. Figures divide the mean estimate by its moment target only where every component estimate is defined; conditional summaries elsewhere retain their inclusion counts. These diagnostics do not isolate each estimation error's causal contribution to rejection.

Variability of the estimated MI variance.

For each selected population, measure the observed variability of $\widehat{V}(P)$ by replacing $\widehat{\Delta}_s$ with $\widehat{V}_s(P)$ in Equation (B.8). This measures how much the variance estimate itself changes between samples. Chapter 4 predicts that variability to first order.

Compare the observed value with

$$\begin{aligned} \text{population prediction} &= \frac{\tau^2(P)}{n_P}, \\ \text{mean plug-in prediction} &= \frac{1}{M}\sum_{s=1}^{M}\frac{\widehat{\tau}_s^2(P)}{n_P}. \end{aligned} \tag{B.9}$$

The population prediction uses the known probabilities, whereas the mean plug-in prediction uses the empirical probabilities in each repetition. Repeat both for Q. For each population, use the same repetitions with finite $\widehat{V}_s$ and $\widehat{\tau}_s^2$, including finite zeros and outputs whose test is invalid. Divide both predictions by the same observed variability. A ratio below one indicates underprediction, a ratio above one indicates overprediction, and one indicates agreement. The observed variability must be positive; a zero prediction gives a zero ratio.

Estimated direction among rejections.

For a nonzero population difference, classify each valid two-sided rejection by the sign of $\widehat{\Delta}_{\mathrm{BC}}$. The estimate has the opposite sign when $\widehat{\Delta}_{\mathrm{BC}}\Delta < 0$; it has the correct sign when $\widehat{\Delta}_{\mathrm{BC}}\Delta > 0$. Report the number of rejections and the number with opposite signs for each method. The fraction with opposite signs divides the opposite-sign count by the rejection count and is undefined if there are no rejections. These counts supplement the estimate of power, which includes every valid rejection regardless of direction.

B.6 Simulation uncertainty

For a rate estimate $\widehat{r}$ from M independent repetitions, its approximate Monte Carlo standard error is $\{\widehat{r}(1-\widehat{r})/M\}^{1/2}$. Rejection and validity rates use pointwise 95% Wilson intervals (Wilson, 1927).

For the paired method contrast, let D_s equal the Expanded Welch valid-rejection indicator minus Normal Wald's valid-rejection indicator in repetition s. The approximate interval for the average paired difference $\bar{D}$ is

$$\bar{D} \;\pm\; 1.96\left\{\frac{1}{M(M-1)}\sum_{s=1}^{M}(D_s-\bar{D})^2\right\}^{1/2}. \tag{B.10}$$

Use the same common-valid subset when reporting a conditional paired contrast. Comparisons across constructions or sample allocations use independent streams and therefore do not use paired indicators. All intervals describe simulation uncertainty at fixed populations, not simultaneous coverage over every setting.

B.7 Component moment targets and uncertainty

The first comparison in Chapter 5 measures the mean and sampling variance of $\widehat{V}(P)$ separately. Expectations and variances refer to repeated samples from the same fixed population at the same sample size. Substituting the predictions in Equation (5.4) into the moment-matching formula gives

$$\nu_{P,\mathrm{FO}} = \frac{2n_P V(P)^2}{\tau^2(P)}. \tag{B.11}$$

This is the population first-order approximation defined in Chapter 4. The sampling-moment target instead uses the actual mean and variance of the estimator:

$$\nu_{P,\mathrm{mom}} = \frac{2\{\mathrm{E}\,\widehat{V}(P)\}^2}{\mathrm{Var}\{\widehat{V}(P)\}}. \tag{B.12}$$

The construction assumes positive $V(P)$ and $\tau^2(P)$, as specified in Chapter 4. The two degrees-of-freedom values can nevertheless differ at finite sample sizes. Dividing their definitions before applying either approximation gives the exact identity

$$\frac{\nu_{P,\mathrm{FO}}}{\nu_{P,\mathrm{mom}}} = \frac{\left\{V(P)/\,\mathrm{E}\,\widehat{V}(P)\right\}^2}{\{\tau^2(P)/n_P\}/\,\mathrm{Var}\{\widehat{V}(P)\}}. \tag{B.13}$$

The numerator is the reciprocal of the squared mean ratio in Figure 6.1; the denominator is the middle-row predicted / observed sampling-variance ratio. A sampling mean $\mathrm{E}\{\widehat{V}(P)\}$ above $V(P)$ lowers the first-order degrees of freedom relative to the moment target when the variance prediction is otherwise accurate. The same calculations are made separately for Q.

Estimating the targets and their uncertainty.

For a component, let $X_s = \widehat{V}_s$ in repetition s. Use all finite values, including zeros, even if component degrees of freedom are undefined. From M repetitions, calculate

$$\bar{X} = \frac{1}{M}\sum_{s=1}^{M} X_s, \qquad S_X^2 = \frac{1}{M-1}\sum_{s=1}^{M}(X_s - \bar{X})^2. \tag{B.14}$$

For $S_X^2 > 0$, the estimated moment target is $2\bar{X}^2/S_X^2$. The diagnostic decomposition uses these evaluation moments.

The delta method uses a first-order Taylor expansion of $f(\mu, v) = 2\mu^2/v$, where μ and v are the sampling mean and variance of $\widehat{V}(P)$. Its derivatives are $\partial f/\partial\mu = 4\mu/v$ and $\partial f/\partial v = -2\mu^2/v^2$. Evaluating them at $\bar{X}, S_X^2$ gives the two coefficients below. The mean and variance errors are propagated jointly because both estimates use

the same repetitions. Define

$$U_s = X_s - \bar{X}, \qquad W_s = (X_s - \bar{X})^2 - S_X^2. \tag{B.15}$$

The first-order contribution to the target's estimation error is

$$Z_s = \frac{4\bar{X}}{S_X^2} U_s - \frac{2\bar{X}^2}{(S_X^2)^2} W_s. \tag{B.16}$$

Its sample SD divided by $\sqrt{M}$ estimates the Monte Carlo standard error. Calculating the SD after combining the two terms retains the mean–variance covariance. The experimental supplement gives the Taylor expansion producing these coefficients.

Mean and variance ratios use the same joint delta-method calculation, keeping the plug-in prediction and observed variance paired. For the fitted null SD, apply the sampling-variance calculation to the corrected MI differences and propagate through the square root, whose derivative at S_X^2 is $1/(2S_X)$.

B.8 Empirical calibration and power uncertainty

Calibration follows the fitting and independent evaluation steps in Chapter 5. For Expanded Welch, assign invalid fitting outputs p-value one and sort the fitting p-values. Set the threshold to the $(\lfloor 0.05M \rfloor + 1)$th smallest value, rejecting only strictly smaller evaluation p-values. Ties at the boundary are excluded. Normal Wald uses the equivalent upper-tail threshold for $|T|$, with invalid fitting values below all valid absolute statistics. This gives at most 5% rejection in the fitting batch; independent null evaluation measures the achieved rate, including discreteness.

The calibration-sensitivity comparison uses the lower and upper nulls in Table 5.4 at $h = 2, 4$ for A–C. For each upper null, solve for the copying probability with the original margin vectors, dimensions and sample sizes. Generate 100,000 independent fitting repetitions and 100,000 validation repetitions for each of these six equal-MI settings. Apply both calibrations to the same original alternatives and to both null validation settings. Invalid outputs remain non-rejections.

All calibrated-power intervals use 1,000 joint bootstrap repetitions. For each full curve, share one lower fitting resample across its points. For the sensitivity comparison, share one lower fitting resample across the two alternatives within each focus setting. Independently resample each upper fitting batch and refit the cutoffs. At each evaluation point, resample the tests' joint decisions; in the sensitivity comparison, retain both calibrations in that joint resample. This preserves dependence between methods and between calibrations when estimating power

differences and changes in those differences. Population tables and effect sizes stay fixed.

Use the 2.5th and 97.5th percentiles for pointwise intervals of rates and paired differences, comparing the latter with the practical tolerance of minus one to plus one percentage point. Cross-null rates use Wilson intervals conditional on the fitted cutoffs, quantifying validation sampling uncertainty only. These checks concern the two specified null settings.

Appendix C

Additional Results

This appendix reports the component diagnostics, calibration sensitivity, rejection directions and multiple-testing adjustment used in the main argument. Each simulation setting uses 100,000 evaluation repetitions. The experimental supplement retains the full companion plots and supporting comparisons.

C.1 Component diagnostics and fitted null SDs

Table C.1 shows how the mean error of $\widehat{V}$ varies across MI values, margin regimes and populations in the 4×4 sweep. The scaled error is $n\{\mathrm{E}\,\widehat{V} - V\}$. The near-independence calculation suggests a mean error of $(k-1)^2/n = 9/n$ for $\widehat{V}$ in these tables. The observed scaled errors vary, so subtracting $9/n$ from $\widehat{V}$ would leave residual mean error.

Table C.1: Scaled mean error across the 4×4 MI sweep.

n	Range of $n\{\mathrm{E}\,\widehat{V} - V\}$
100	5.55–8.45
1,000	7.50–9.39
10,000	7.60–10.19

At $n = 100$, the whole scaled-error range is below nine, so a $9/n$ subtraction would overcorrect the mean throughout the checked grid. At the two larger sample sizes the ranges straddle nine: the same correction would be too large for some populations and too small for others. The wider scaled ranges do not imply larger absolute errors, since the entries must be divided by n to recover the mean error of $\widehat{V}$. They show that its coefficient depends on MI and margins.

Table C.2 gives the means, medians and fifth–95th percentile ranges of component degrees of freedom. The moment target uses the evaluation sampling moments. All 100,000 component estimates are positive and finite in each row; percentile ranges describe sampling spread.

Table C.2: Sampling spread of component degrees of freedom.

Setting, population	Mean df	Median df	5th–95th percentiles	Moment target	Included
A, P	6.52	5.23	[1.95, 15.32]	14.81	100,000
A, Q	6.54	5.24	[1.93, 15.16]	14.87	100,000
B, P	14.04	12.74	[5.83, 26.60]	23.54	100,000
B, Q	9.72	8.47	[2.64, 20.91]	12.47	100,000
C, P	157.09	156.66	[131.26, 184.27]	157.96	100,000
C, Q	139.70	139.28	[115.88, 165.12]	141.27	100,000
16×16, same skew $n = 1{,}000$, P	39.78	39.32	[29.18, 51.92]	71.67	100,000
16×16, same skew $n = 1{,}000$, Q	39.80	39.35	[29.33, 51.88]	72.18	100,000
16×16, same skew $n = 10{,}000$, P	133.18	132.82	[114.30, 153.25]	148.48	100,000
16×16, same skew $n = 10{,}000$, Q	133.17	132.92	[114.19, 153.21]	146.84	100,000
16×16, different skew $n = 1{,}000$, P	52.58	52.08	[41.98, 64.93]	87.28	100,000
16×16, different skew $n = 1{,}000$, Q	39.80	39.33	[29.15, 51.94]	71.05	100,000
16×16, different skew $n = 10{,}000$, P	172.67	172.41	[152.55, 193.72]	189.98	100,000
16×16, different skew $n = 10{,}000$, Q	133.24	132.99	[114.21, 153.33]	147.07	100,000

In Table C.2, the means at A and B exceed the medians and the upper percentile extends farther from the median than the lower percentile. Large estimates in the upper tail therefore affect the mean, which alone does not describe a typical repetition. Both summaries remain below the moment targets. At C, the mean and median nearly coincide and the percentile range is much narrower relative to their size. The much larger component degrees of freedom are also more stable in relative terms, although their absolute percentile ranges are wider.

The P and Q rows of Table C.2 nearly match at A and the same-skew settings, as expected for identical population tables. Under different skew, Q has lower estimated component degrees of freedom and a lower moment target. Increasing the 16×16 sample size raises the estimates, narrows their relative spread and brings their means closer to the targets in both populations. The remaining gaps at the smaller sample size occur with complete inclusion, so they are not produced by dropping invalid component estimates.

Table C.3 describes the shared difference and standard error. Bias/SD divides difference bias by its observed SD; variance ratio divides the mean squared standard error by the observed difference variance. Rank corr. is Spearman's correlation between $|\widehat{\Delta}_{\mathrm{BC}} - \mathrm{mean}(\widehat{\Delta}_{\mathrm{BC}})|$ and $\widehat{\mathrm{SE}}$, using every repetition where both quantities are finite without a test-validity restriction. Skewness measures asymmetry in $\widehat{\Delta}_{\mathrm{BC}}$: positive values indicate a longer right tail and negative values a longer left tail. "Median combined df" summarises $\widehat{\nu}_{\mathrm{expanded}}$, rather than either component's degrees of freedom. All 100,000 repetitions are included in each row.

A near-zero signed correlation can follow from symmetry despite magnitude dependence: the same-skew 2×2, $n = 100$ null has Pearson correlation between $\widehat{\Delta}_{\mathrm{BC}}$ and $\widehat{\mathrm{SE}}$ about -0.001, but rank correlation between the centred magnitude and $\widehat{\mathrm{SE}}$ of 0.547.

Table C.3: Shared-estimator diagnostics and combined degrees of freedom.

Setting	Bias/SD	Est./obs. variance	Rank corr.	Skewness	Median combined df
A	0.003	0.827	0.017	0.002	10.61
B	0.870	1.535	0.122	-0.052	21.12
C	0.000	1.010	0.060	-0.013	293.23
16×16, same skew $n = 1{,}000$	-0.003	1.929	0.106	0.008	78.87
16×16, same skew $n = 10{,}000$	0.001	1.047	0.066	0.012	265.85
16×16, different skew $n = 1{,}000$	2.058	1.938	0.114	-0.008	88.51
16×16, different skew $n = 10{,}000$	-0.229	1.064	0.055	-0.013	299.21

The small skewness values in Table C.3 concern symmetry about the sampling mean, whereas Bias/SD concerns displacement from the true difference. The different-skew sparse case can therefore be approximately symmetric while centred more than two

observed SDs above zero. Its median combined degrees of freedom are already large enough for the two rejection cutoffs to be fairly close, yet both tests fail because the shared statistic remains displaced.

Across the sample-size rows of Table C.3, increasing n from 1,000 to 10,000 gives both sparse regimes variance ratios much closer to one, weaker magnitude dependence and higher combined degrees of freedom. The different-skew bias also falls sharply and changes sign. These changes accompany the recovery of the false-positive rates in Table 6.4. The modest rank correlations in these selected rows differ from the much stronger 2×2 dependence noted above; one average variance ratio cannot describe all these sampling distributions.

Additional checks examine the peaks in Figure 6.1. For 16×16 different-skew populations at common MI 0.02, the cell contributions to $\widehat{V}$ and $\widehat{\tau}^2$ are partitioned into cells observed once and cells observed more than once. The check uses 10,000 evaluation repetitions at each $n \in \{10, 20, 50, 100, 250, 1000, 10000\}$, for both populations. Once-observed cells account for most of the mean estimates around the peak, but much less at the largest checked sample size. Comparing the linear change derived in Chapter 4 with $\widehat{V}(P) - V(P)$ also shows substantial remaining variation around the peak. Its covariance with the linear change is retained when checking the total sampling variance.

For the MI sweep in Figure 6.3 (4×4, different skew, $n = 1{,}000$), the local approximations $\widehat{V} \approx 2\widehat{I}$ and $\widehat{\tau}^2 \approx 4\widehat{V}$ are checked against the sampled values. Twice the plug-in MI increasingly understates the mean and sampling variance of $\widehat{V}$, while $4\widehat{V}$ increasingly understates the mean of $\widehat{\tau}^2$. The observed variance of $2n\widehat{I}$ also exceeds its local noncentral chi-squared prediction. Both parts of the local calculation therefore contribute to the separation in Figure 6.3.

The uniform 2×2 panel in Figure 6.4 shows a further source of conservatism: the centred magnitude of $\widehat{\Delta}_{\mathrm{BC}}$ is positively associated with $\widehat{\mathrm{SE}}$, especially around the rejection-rate trough. The magnitude correlation falls from 0.656 at $n = 50$ to 0.167 at $n = 1{,}000$. This check uses the same correlation defined above; it concerns the joint behaviour of the estimates, which their average variance ratio alone does not describe.

The null SDs in Table C.4 use independent 100,000-repetition fitting batches. Their Monte Carlo standard errors follow Appendix B.7. The fitted SDs define the fixed alternative populations, so power intervals condition on those tables.

Table C.4: Independently fitted null standard deviations.

Setting	Null SD (nats)	Monte Carlo SE (nats)
A	0.154967	0.000363
B	0.065138	0.000152
C	0.003175	0.000007

The fitted SD is largest at A and smallest at C, so a common number of null SDs corresponds to very different MI differences in nats. Each Monte Carlo SE is small relative to its fitted SD, showing that these scales are estimated precisely within the specified fitting populations.

C.2 Calibrated power and sensitivity checks

Figure 6.7 in Chapter 6 shows the full power curves under calibration at common MI 0.02. Table C.5 gives nominal and calibrated paired contrasts at two null SDs, including cutoff-fitting uncertainty for the calibrated intervals.

Table C.5: Expanded Welch minus Normal Wald power at two null SDs, in percentage points.

Setting	Nominal	95% interval	Calibrated	95% interval
A	-5.128	[-5.265, -4.991]	+3.580	[+3.295, +3.787]
B	-0.727	[-0.780, -0.674]	+1.112	[+1.028, +1.194]
C	-0.304	[-0.338, -0.270]	+0.044	[+0.013, +0.071]

In Table C.5, the nominal power loss is largest at A and much smaller at B and C. After calibration, the ordering reverses at A and B, while C's gain remains small. Its interval excludes zero but stays well within the one-percentage-point tolerance: a precisely estimated difference can still be practically small. The following tables examine whether the larger A/B gains persist under a different fitting null.

Along the alternative paths in Figure 6.6, the sampling SD also changes. At four fitted null SDs, the observed SD of $\widehat{\Delta}_{\mathrm{BC}}$ is approximately $1.41\sigma_0$ at A, $1.70\sigma_0$ at B and $1.16\sigma_0$ at C. Difference bias remains positive at A and B but negligible at C. These checks support the cross-panel interpretation of Figure 6.6: a common null-SD separation does not give the same distance from zero relative to the alternative sampling variation.

Table C.6 compares both calibrations on the same fixed alternatives. The lower null has common MI 0.02; the upper null sets both MI values to the alternative's larger value. Appendix B.8 specifies the construction and joint bootstrap. Table 6.6 in Chapter 6 gives Expanded Welch minus Normal Wald power differences with intervals.

Table C.6: Power for fixed alternatives under lower- and upper-null calibration.

Setting, h, fitting null	Normal Wald	Expanded
A: 2, lower	32.57%	36.15%
A: 2, upper	18.58%	17.83%
A: 4, lower	73.84%	77.84%
A: 4, upper	54.91%	55.09%
B: 2, lower	4.42%	5.54%
B: 2, upper	4.68%	4.73%
B: 4, lower	27.66%	31.59%
B: 4, upper	26.40%	26.01%
C: 2, lower	45.61%	45.66%
C: 2, upper	45.47%	45.42%
C: 4, lower	93.56%	93.58%
C: 4, upper	93.42%	93.41%

Across the two- and four-null-SD rows of Table C.6, power increases for both tests under each calibration, but changing the fitting null has different effects across settings. At A, upper-null fitting substantially lowers both power estimates at either separation. At B, the changes are smaller and method-specific: at two SDs, Wald's power rises slightly while Expanded Welch's falls. At C, both power levels and method differences are nearly unchanged by the fitting null. Because each alternative sample is held fixed between calibrations, these changes come from the fitted thresholds.

Table C.7 applies each fitted cutoff to both null validation settings. Each row includes all 100,000 repetitions, with invalid outputs contributing no rejection. The two null settings distinguish performance where thresholds were fitted from performance at a different common MI.

Table C.7: Independent null validation under both fitted calibrations.

		Lower evaluation null		Upper evaluation null	
Setting, h	Fitting null	Normal Wald	Expanded	Normal Wald	Expanded
A: 2	lower	4.93%	4.99%	11.20%	13.71%
A: 2	upper	1.63%	0.79%	4.87%	4.84%
A: 4	lower	4.93%	4.99%	12.56%	15.58%
A: 4	upper	1.34%	0.53%	5.04%	5.00%
B: 2	lower	4.98%	4.96%	4.82%	5.76%
B: 2	upper	5.26%	4.16%	5.04%	5.04%
B: 4	lower	4.98%	4.96%	5.37%	6.76%
B: 4	upper	4.56%	3.27%	4.99%	4.99%
C: 2	lower	4.98%	4.98%	4.97%	4.99%
C: 2	upper	4.93%	4.90%	4.94%	4.94%
C: 4	lower	4.98%	4.98%	5.03%	5.06%
C: 4	upper	4.84%	4.81%	4.87%	4.86%

In Table C.7, lower-null calibration at A and $h = 2$ gives rates close to 5% at the lower evaluation null, but more than twice that target at the upper null. Upper-null fitting reverses the pattern: rejection is close to 5% there and conservative at the lower null. The same pattern is stronger at $h = 4$, where the two common MI values are farther apart.

At B, transfer errors are smaller than at A but differ between methods. With lower-null fitting, Expanded Welch has higher rejection than Wald at the upper null and exceeds 5%; with upper-null fitting, it becomes more conservative at the lower null. These differences grow at the larger separation. At C, rates remain near 5% across both fitting choices, evaluation nulls and separations, consistent with the stable power comparison. The sensitivity of power at A and B therefore accompanies changes in their relative false-positive rates when calibration is transferred. Neither calibration establishes control across the complete equal-MI null.

C.3 Sparse-table bias and estimated direction

Table C.8 counts wrong-direction rejections at 16×16, $n_P = n_Q = 1{,}000$ and $I(P) - I(Q) = -0.01$ nats. Each entry gives the number of wrong-direction rejections divided by the total number of rejections. Almost every rejection under

different-skew margins indicates the wrong ordering; the same-skew populations show far fewer such cases.

Table C.8: Wrong-direction counts among rejections at the negative MI difference.

Method	Different skew	Same skew
Normal Wald	10,183 / 10,184	59 / 1,809
Expanded Welch	9,591 / 9,592	52 / 1,646

Expanded Welch reduces the number of rejections in both columns, but under different skew almost every retained rejection still has the wrong direction. Its larger cutoff removes some rejections without repairing the sign error in the estimated difference.

At $n = 10{,}000$, neither test has a wrong-direction rejection at this alternative under either skew regime. Under different skew, the difference bias at $n = 1{,}000$ is 0.03071 nats. Thus

$$\mathrm{E}(\widehat{\Delta}_{\mathrm{BC}}) \approx -0.01 + 0.03071 = 0.02071 \text{ nats.} \tag{C.1}$$

The positive centre explains why almost every rejection indicates the wrong ordering. Two-sided power retains every rejection; these counts show why its interpretation also requires the estimated direction.

C.4 Holm adjustment for the application

Table C.9 gives raw and Holm-adjusted p-values for six tests of equal predictor–outcome MI between groups. Adult compares female and male records using education (16 categories) or occupation (14 categories) and income above US$50,000 (Becker and Kohavi, 1996). Its training and test partitions are combined without survey weights; records with missing occupation are excluded only from that comparison. Student Performance compares schools GP and MS using Portuguese-language course records, study time (four levels) or maternal education (five levels), and a final grade of at least 10 (Cortez, 2008). Shopping compares new and returning visitors for month–purchase and weekend–purchase MI.

Table C.9: Raw and Holm-adjusted p-values for the six exploratory comparisons.

	Normal Wald		Expanded Welch	
Comparison	Raw	Adjusted	Raw	Adjusted
Adult: occupation	1.18×10^{-44}	7.10×10^{-44}	1.64×10^{-42}	9.81×10^{-42}
Adult: education	1.89×10^{-34}	9.45×10^{-34}	4.93×10^{-33}	2.46×10^{-32}
Online Shoppers: month	4.71×10^{-5}	1.89×10^{-4}	3.73×10^{-4}	1.49×10^{-3}
Student Performance: study time	0.1982	0.5946	0.2339	0.7016
Student Performance: maternal education	0.8361	1	0.8468	1
Online Shoppers: weekend	0.9554	1	0.9601	1

Apply Holm's adjustment separately to each method's six p-values (Holm, 1979). Order the raw p-values and multiply them by 6, 5, 4, 3, 2 and 1. Moving through that order, replace any value below the preceding adjusted value with the preceding adjusted value, and cap the results at one.

Both Adult comparisons and the shopping month comparison remain below 0.05 after adjustment for either method. The other three comparisons remain above 0.05. Expanded Welch gives larger raw p-values throughout the table, but it selects the same three comparisons as Normal Wald at this threshold. Holm's adjustment changes the reported p-values without changing those decisions in this exploratory family.

Bibliography

Evan Archer, Il Memming Park, and Jonathan W. Pillow. Bayesian and quasi-Bayesian estimators for mutual information from discrete data. *Entropy*, 15(5): 1738–1755, 2013. doi: 10.3390/e15051738.

Moulinath Banerjee. Likelihood ratio tests under local alternatives in regular semiparametric models. *Statistica Sinica*, 15(3):635–644, 2005. URL https://www3.stat.sinica.edu.tw/statistica/J15N3/J15N33/J15N33.html.

Barry Becker and Ronny Kohavi. Adult. UCI Machine Learning Repository, 1996. URL https://doi.org/10.24432/C5XW20. Dataset.

Robert M. Bell and Daniel F. McCaffrey. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*, 28(2):169–181, 2002. URL https://www150.statcan.gc.ca/n1/pub/12-001-x/2002002/article/9058-eng.pdf.

Cristiano Bocci, Enrico Carlini, and Fabio Rapallo. Geometry of diagonal-effect models for contingency tables. In Marlos A. G. Viana and Henry P. Wynn, editors, *Algebraic Methods in Statistics and Probability II*, volume 516 of *Contemporary Mathematics*, pages 61–73. American Mathematical Society, Providence, Rhode Island, 2010. doi: 10.1090/conm/516/10166. URL https://arxiv.org/abs/0908.0232.

Terry Bossomaier, Lionel Barnett, Michael Harré, and Joseph T. Lizier. *An Introduction to Transfer Entropy: Information Flow in Complex Systems*. Springer, Cham, 2016. doi: 10.1007/978-3-319-43222-9.

Richard P. Brent. *Algorithms for Minimization without Derivatives*. Prentice-Hall, Englewood Cliffs, New Jersey, 1973.

David R. Brillinger. Some data analyses using mutual information. *Brazilian Journal of Probability and Statistics*, 18(2):163–182, 2004. URL https://www.stat.berkeley.edu/~brill/Papers/MIBJPS.pdf.

Anne Chao and Tsung-Jen Shen. Nonparametric estimation of Shannon's index of diversity when there are unseen species in sample. *Environmental and Ecological Statistics*, 10(4):429–443, 2003. doi: 10.1023/A:1026096204727.

EunYi Chung and Joseph P. Romano. Exact and asymptotically robust permutation tests. *Annals of Statistics*, 41(2):484–507, 2013. doi: 10.1214/13-AOS1090.

Paulo Cortez. Student Performance. UCI Machine Learning Repository, 2008. URL https://doi.org/10.24432/C5TG7T. Dataset.

Robert Drake and Apratim Guha. A mutual information-based k-sample test for discrete distributions. *Journal of Applied Statistics*, 41(9):2011–2027, 2014. doi: 10.1080/02664763.2014.899325.

Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.

Michael P. Fay and Barry I. Graubard. Small-sample adjustments for Wald-type tests using sandwich estimators. *Biometrics*, 57(4):1198–1206, 2001. doi: 10.1111/j.0006-341X.2001.01198.x.

Apratim Guha and Tom Chothia. A two sample test based on mutual information. *Calcutta Statistical Association Bulletin*, 66(1–2):39–54, 2014. doi: 10.1177/0008068320140103.

Frank R. Hampel. The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 69(346):383–393, 1974. doi: 10.1080/01621459.1974.10482962.

Damián G. Hernández and Inés Samengo. Estimating the mutual information between two discrete, asymmetric variables with limited samples. *Entropy*, 21(6):623, 2019. doi: 10.3390/e21060623.

Sture Holm. A simple sequentially rejective multiple test procedure. *Scandinavian Journal of Statistics*, 6(2):65–70, 1979. URL https://www.jstor.org/stable/4615733.

Kermit Hutcheson. A test for comparing diversities based on the Shannon formula. *Journal of Theoretical Biology*, 29(1):151–154, 1970. doi: 10.1016/0022-5193(70)90124-4.

Marcus Hutter. Distribution of mutual information. In *Advances in Neural Information Processing Systems 14*, pages 399–406, 2002. Conference held in 2001; arXiv:cs/0112019.

Marcus Hutter and Marco Zaffalon. Distribution of mutual information from complete and incomplete data. *Computational Statistics & Data Analysis*, 48(3): 633–657, 2005. doi: 10.1016/j.csda.2004.03.010.

Guido W. Imbens and Michal Kolesár. Robust standard errors in small samples: Some practical advice. *The Review of Economics and Statistics*, 98(4):701–712, 2016. doi: 10.1162/REST_a_00552.

Arnold Janssen. Studentized permutation tests for non-i.i.d. hypotheses and the generalized Behrens–Fisher problem. *Statistics & Probability Letters*, 36(1):9–21, 1997. doi: 10.1016/S0167-7152(97)00043-6.

Kirthevasan Kandasamy, Akshay Krishnamurthy, Barnabas Poczos, Larry Wasserman, and James M. Robins. Nonparametric von Mises estimators for entropies, divergences and mutual informations. In *Advances in Neural Information Processing Systems 28*, pages 397–405, 2015.

Göran Kauermann and Raymond J. Carroll. A note on the efficiency of sandwich covariance matrix estimation. *Journal of the American Statistical Association*, 96 (456):1387–1396, 2001. doi: 10.1198/016214501753382309.

Marius Marinescu and Costel Balcau. On the use of mutual information for testing independence, 2025. arXiv:2502.17636.

George A. Miller. Note on the bias of information estimates. In Henry Quastler, editor, *Information Theory in Psychology: Problems and Methods*, pages 95–100. Free Press, Glencoe, Illinois, 1955.

Roelof Moddemeijer. On estimation of entropy and mutual information of continuous distributions. *Signal Processing*, 16(3):233–248, 1989. doi: 10.1016/0165-1684(89)90132-1.

Roelof Moddemeijer. A statistic to estimate the variance of the histogram-based mutual information estimator based on dependent pairs of observations. *Signal Processing*, 75(1):51–63, 1999. doi: 10.1016/S0165-1684(98)00224-2.

Ricardo Mora and Javier Ruiz-Castillo. The statistical properties of the mutual information index of multigroup segregation. Working Paper, Economic Series (48) 09-84, Universidad Carlos III de Madrid, Departamento de Economía, 2009.

Ricardo Mora and Javier Ruiz-Castillo. Entropy-based segregation indices. *Sociological Methodology*, 41(1):159–194, 2011. doi: 10.1111/j.1467-9531.2011.01237.x.

Tim P. Morris, Ian R. White, and Michael J. Crowther. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*, 38(11):2074–2102, 2019. doi: 10.1002/sim.8086.

Ilya Nemenman, Fariel Shafee, and William Bialek. Entropy and inference, revisited. In Thomas G. Dietterich, Suzanna Becker, and Zoubin Ghahramani, editors, *Advances in Neural Information Processing Systems*, volume 14, pages 471–478. MIT Press, 2002. URL https://papers.neurips.cc/paper_files/paper/2001/file/d46e1fcf4c07ce4a69ee07e4134bcef1-Paper.pdf.

Wei Pan and Melanie M. Wall. Small-sample adjustments in using the sandwich variance estimator in generalized estimating equations. *Statistics in Medicine*, 21 (10):1429–1441, 2002. doi: 10.1002/sim.1142.

Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191–1253, 2003. doi: 10.1162/089976603321780272.

Stefano Panzeri and Alessandro Treves. Analytical estimates of limited sampling biases in different information measures. *Network: Computation in Neural Systems*, 7(1):87–107, 1996. doi: 10.1088/0954-898X/7/1/006.

Angelo Piga, Lluc Font-Pomarol, Marta Sales-Pardo, and Roger Guimerà. Bayesian estimation of information-theoretic metrics for sparsely sampled distributions. *Chaos, Solitons & Fractals*, 180:114564, 2024. doi: 10.1016/j.chaos.2024.114564.

Mark S. Roulston. Estimating the errors on measured entropy and mutual information. *Physica D: Nonlinear Phenomena*, 125(3–4):285–294, 1999. doi: 10.1016/S0167-2789(98)00269-3.

C. Sakar and Yomi Kastro. Online Shoppers Purchasing Intention Dataset. UCI Machine Learning Repository, 2018. URL https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset. Dataset.

Franklin E. Satterthwaite. An approximate distribution of estimates of variance components. *Biometrics Bulletin*, 2(6):110–114, 1946. doi: 10.2307/3002019.

Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948. Parts I and II.

Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, Cambridge, 1998. doi: 10.1017/CBO9780511802256.

Abraham Wald. Tests of statistical hypotheses concerning several parameters when the number of observations is large. *Transactions of the American Mathematical Society*, 54(3):426–482, 1943. doi: 10.1090/S0002-9947-1943-0012401-3.

Bernard L. Welch. The generalization of "Student's" problem when several different population variances are involved. *Biometrika*, 34(1–2):28–35, 1947. doi: 10.1093/biomet/34.1-2.28.

Samuel S. Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1):60–62, 1938. doi: 10.1214/aoms/1177732360.

Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953.